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A stochastic queueing model for travel time reliability with spatial density heterogeneity

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ABSTRACT

This paper constructs a theoretical model connecting day-to-day travel time reliability (TTR) and vehicle density, testing it on freeway stretches in Minneapolis - St. Paul and San Diego. We establish the presence of a counter-clockwise hysteresis loop between average vehicle density and experienced travel time standard deviation. We also observe that vehicle density heterogeneity exerts a more profound effect on TTR during moderate congestion. We find that a time-homogeneous Poisson process accurately characterizes the vehicle arrival process on segments of freeway stretches, both with and without on-ramps, during the morning peak period. The travel times for these segments align well with an exponential distribution. Through an analysis of the travel time correlations among different segments of the freeway stretch, we find that this correlation is higher during congestion offset periods, resulting in more severe travel time fluctuations due to increased density heterogeneity. Using the M/M/1 queueing model, we derive a model encapsulating the variance of experienced travel time, average vehicle density, and density heterogeneity for freeway stretches consisting of multiple segments. We use the 'birth-death process' to model the relationship between average vehicle density and density heterogeneity. By integrating these two models, we derive a novel model that accurately approximates freeway stretch TTR based on average vehicle density and traffic flow. We apply our models to TTR changes on the two selected freeway stretches to validate across different time periods and select two subnetworks, each containing three additional freeway stretches, to validate across spatial dimensions. The results from these experiments corroborate the robustness and generality of our model across both temporal and spatial contexts.

1. Introduction

Travel time unpredictability originates from variability in both the conditions on road networks and traffic demand. Travel Time Reliability (TTR) influences travel decisions and is widely factored into route selection as highlighted by Levinson and Zhu (2013), Brownstone and Small (2005), Zang et al. (2022). Travelers generally prefer reliable routes with slightly longer expected travel times over less reliable routes with lower expected travel times. Furthermore, TTR affects the departure time, chosen mode of travel, and other decisions, with these choices being influenced by the perceived Value of Travel Time Savings (VOT) and the Value of Travel Time Reliability (VOR) (Carrion and Levinson, 2012). Adjusting departure time (leaving earlier) is typically the simplest response to

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unpredictable travel time (Li et al., 2010). Accurately capturing this empirical relationship between TTR and departure time would lead to better predictions of TTR.

Bates et al. (1987) subdivided TTR into three categories: vehicle-to-vehicle reliability, which considers variability among different vehicles using the same route simultaneously; period-to-period reliability, accounting for variations in travel times for vehicles using the same route at different times within a day; and day-to-day reliability, which covers the fluctuations in travel time for vehicles traversing the same route during the same period across different days (Noland and Polak, 2002).

Bates et al. (2001) indicated that day-to-day reliability is essentially random. Once systematic changes like seasonal and day-of-week effects are considered, traffic incidents become the major cause of unreliability. Following this, much of the TTR research has focused on this day-to-day variability and the stochastic nature of TTR (Hollander, 2006; Clark and Watling, 2005; Torrisi et al., 2017; Bates et al., 2003). In addition, empirical studies have shown a decrease in TTR with increasing congestion severity, which results in a counter-clockwise hysteresis loop, suggesting a degree of regularity in TTR (Torrisi et al., 2017; Emam and Al-Deek, 2006; Yildirimoglu et al., 2015). Mahmassani et al. (2013) linked TTR and traffic flow at the network level, indicating that dispersion of the travel time increases with the increasing value of the mean travel time linearly, and the TTR of the network will deteriorate seriously when the traffic flow exceeds its maximum network flow rate and starts to decline. Fosgerau (2010) adopted the bottleneck model (Vickrey, 1969) to describe congestion in terms of dynamic queues, highlighting how expected delays and delay variances fluctuate over time.

However, a connection between hysteresis and the heterogeneity of vehicle density distribution has yet to be established, despite comprehensive research into density heterogeneity (Saedi et al., 2020; Ramezani et al., 2015; Han et al., 2020; Geroliminis and Sun, 2011a; Saberi and Mahmassani, 2012).

Several factors contribute to day-to-day TTR, including natural factors such as weather conditions (Mathew and Pulugurtha, 2022; Zhang and Chen, 2019) and natural disasters (Koetse and Rietveld, 2009), as well as man-made factors like accidents, work zones, and bottlenecks (Gao and Levinson, 2023). Among these, recurring bottlenecks are a prominent cause of unreliability, particularly during morning peak travel times, which are dominated by routine work and school trips (Kwon et al., 2011; Gao and Levinson, 2024a; Parthasarathi et al., 2011).

While queueing models are widely used to describe bottlenecks on freeways to predict traffic flow or travel time, they have not yet been used to describe TTR (Newell, 1993; Li et al., 2013; Cheng et al., 2022). Gao and Levinson (2021) discovered a temporal correlation between Travel Time Reliability (TTR) and density heterogeneity, specifically occupancy heterogeneity. However, that study was primarily focused on the day-to-day reliability of instantaneous travel time within freeway sub-networks.

Extending our previous work on multi-stage spatial queueing models (Gao and Levinson, 2024b), this paper introduces stochastic queueing theory-based measures for day-to-day travel time reliability. We delve deeper into the relationship between spatial effects, specifically the heterogeneity of congestion across freeway segments, and TTR. We identify a deterministic relationship between these factors, which can guide future traffic control strategies and transport policies aimed at enhancing freeway TTR.

The remainder of this paper is organized as follows: Section 2.1 describes the scope of data collection and filtering methods for selected freeway stretches; Section 2.2 explores the hysteresis phenomenon on the freeway stretch; Section 3 describes how the M/M/1 model can be used to describe the vehicle arrival and travel time distribution, generalizes the model to the freeway stretch level and defines the mathematical form; Section 4.1 statistically estimates the model under different conditions; Section 4.2 validates the model across time and space; Section 5 concludes the paper.

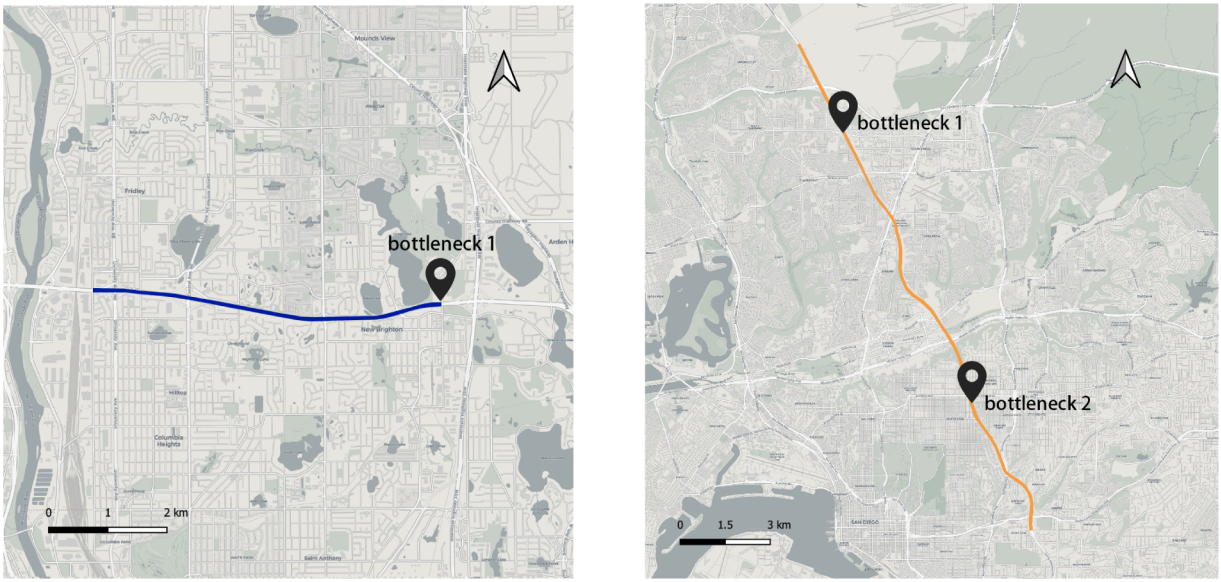
2. Data

Travel time reliability (TTR) is defined as the probability of a trip being successfully completed within a specified time duration (Rakha et al., 2010). The traditional TTR measurement can be divided into mean-variance and scheduling delay approaches (Carrion and Levinson, 2012; Bates et al., 2001). This paper uses the mean-variance approach and measures standard deviation (STD) (Sterman and Schofer, 1976; Van Lint et al., 2008) to indicate the TTR for the selected freeway stretches. Given previous studies (Geroliminis and Sun, 2011a; Saberi and Mahmassani, 2012) on the impact of density heterogeneity on traffic flow throughput capacity, we hypothesize that it has a similar effect on TTR. Therefore, we will explore the impact of density heterogeneity on TTR from an empirical perspective first.

2.1. Data sources

Fig. 1 shows the selected freeway stretches from Minneapolis - St. Paul (MSP) and San Diego. On the MSP freeway network, one shorter freeway stretch, I-694 (W-E), with a total length of around 5.4 km, is selected. The freeway stretch has 3 lanes and 9 loop detectors, which are approximately evenly distributed on the mainline. The average occupancy of the freeway stretch during the morning peak period is 13.53%. On the San Diego freeway network, one longer freeway stretch I-805 (S-N) is selected with a total length of 18.1 km. The freeway stretch has 4 lanes and contains 15 approximately evenly distributed loop detectors. The average occupancy during the morning peak period is 24%.

For data collection, limited by the precision of the data source, we select 1 min loop detector data and 5 min loop detector data for all working days in 2018 during the morning peak period from 6:00 am to 10:00 am for the MSP and San Diego freeway networks, which are obtained from Mn/DOT Traffic Data (Minnesota Department of Transportation, 2022) and California freeway Performance Measurement System (PeMS) (California Department of Transportation, 2021), respectively. Because we want to explore the impact of spatial distribution of recurrent congestion on TTR, we need to exclude the impact of weekends and holidays, as well as the adverse



(a) I-694 (W-E) from Minneapolis - St. Paul network (b) I-805 (S-N) from San Diego network

Fig. 1. Selected stretches from different freeway networks.

effects of extreme weather and traffic crashes on traffic conditions, which are taken into account as outliers are removed if they exceed 3 standard deviations from the mean.

In addition, we also need to ensure that no congestion occurred upstream and downstream of the freeway stretch to avoid exogenous factors leading to the fluctuation of TTR. This external congestion is identified as times when the average speed of the upstream and downstream loop detectors during the morning peak period are lower than 40 km/h (Chiabaut and Faitout, 2021). Therefore, after filtering the data in 2018, there are 194 days and 201 days of data that meet these requirements for the MSP and San Diego freeway networks, respectively. After analyzing the speed data on these congested days, for freeway stretch I-694 (W-E), there is a recurrent bottleneck at the exit. For freeway stretch I-805 (S-N), there are two recurrent bottlenecks located 3.2 km from the exit and 4.1 km from the entrance, respectively.

2.2. Data exploration: freeway stretch TTR and density heterogeneity

To determine the TTR of the freeway stretch, we first need to determine the travel time of the segment based on the loop detector it contains. It is well known that the measured travel time based on the loop detector data will cause serious underestimation compared with the real travel time (Li et al., 2006; Yildirimoglu and Geroliminis, 2013) due to acceleration and deceleration, reaction time, shock-waves, and other factors. Unlike the traditional constant speed interpolation (CSI) method (Yildirimoglu et al., 2015; Mori et al., 2015; Soriguera and Robuste, 2010) to calculate the travel time, which calculates the travel speed of a segment by averaging the speed readings at two consecutive detectors, this paper proposes a half-segment (HS) method to calculate the segment travel time, which assumes that each detector covers only half the distance of its adjacent segments and can offset the underestimation of the travel time well, as detailed shown in Appendix A.

The travel time of the freeway stretch is considered as the experienced travel time (Yildirimoglu and Geroliminis, 2013; Mark et al., 2004), which means that the vehicle travel time is obtained by passing through each mainline loop detector on the freeway stretch in turn, and this method can more truly reflect the continuous travel time that driving through the entire freeway stretch. The heterogeneity of the spatial distribution of congestion refers to the sum of the mean squared errors between vehicle density of each segment with its average value of all segments on the freeway stretch, and after day-to-day averaging, which can well reflect the extent of the heterogeneity of congestion on the freeway stretch. Note that the density cannot be directly measured by the loop detector but is estimated based on the conversion relationship between occupancy and density (Minnesota Department of Transportation, 2022; California Department of Transportation, 2021), and since most of the morning peak period is in congested conditions, the errors caused by this estimation method can be treated as systematic bias. The formulas for the experienced travel time mean and STD, as well as the vehicle density mean and heterogeneity are shown in Appendix B. Therefore, the relationship between average vehicle density and experienced travel time STD and vehicle density heterogeneity for selected freeway stretches I-694 and I-805 is shown in Fig. 2.

It should be noted that we normalize the experienced travel time mean and STD to the value in min/km (i.e. pace) to facilitate the comparison of the results of the two selected freeway stretches. We observe (Fig. 2) there are counter-clockwise hysteresis loops in the average vehicle density and experienced travel time mean, STD and vehicle density heterogeneity curves, and are more significant for travel time STD and density heterogeneity. Furthermore, the value of experienced travel time STD and vehicle density heterogeneity

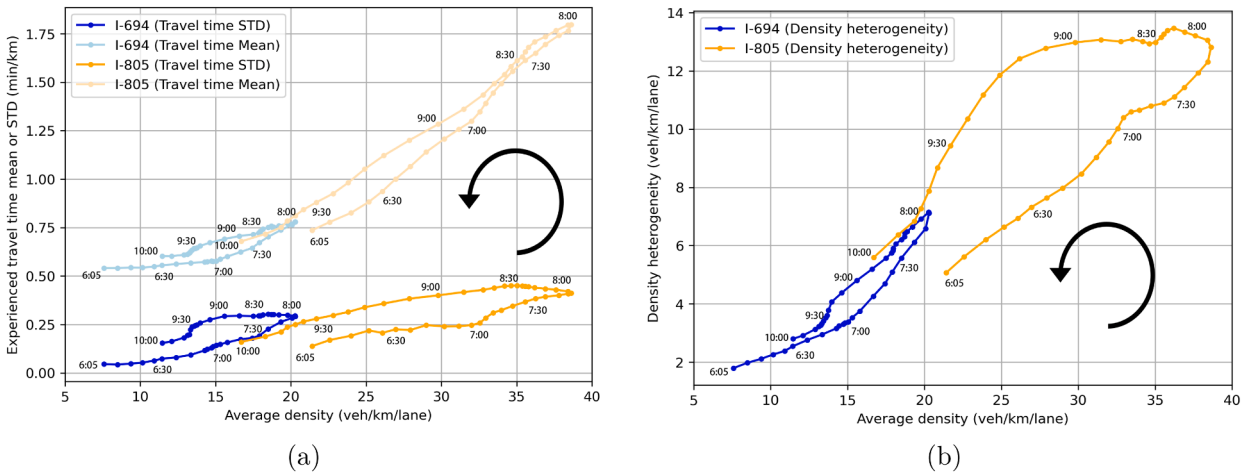


Fig. 2. Relationship between average vehicle density and experienced travel time mean and STD (a) and average vehicle density and vehicle density heterogeneity (b) for selected freeway stretches.

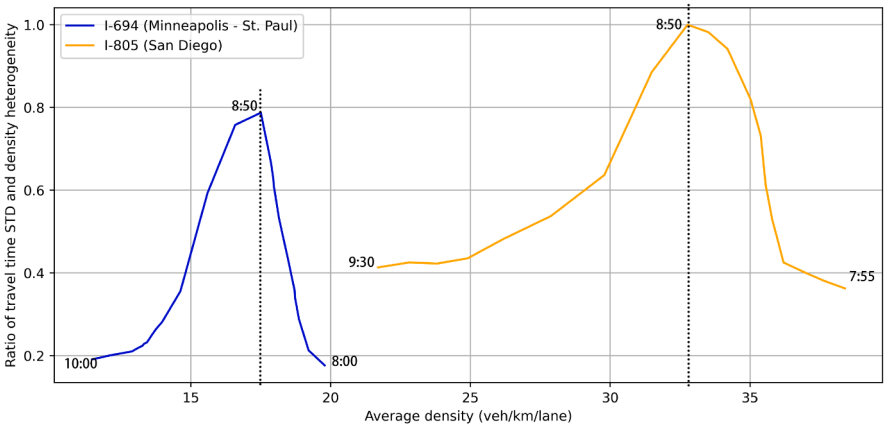


Fig. 3. Diurnal curves of ratio between the difference of travel time STD and the difference of density heterogeneity.

has a strong correlation at the same departure time interval, this phenomenon indicates that the spatial distribution of congestion also has regularity effects on TTR. In addition, for both freeway stretches, we notice that the travel time STD curves all form a shuttle-like shape (narrow on both sides and wide in the middle). We hypothesize that when the average density is the same, the density heterogeneity affects the TTR most seriously when the average density is close to the value of congestion onset start and offset end.

To test our hypothesis, during the morning peak period, we select the time interval pairs from congestion onset and offset, where the average density intersects to keep the selected time interval pairs under the same average density. We determine the effect of density heterogeneity on travel time STD by the ratio between the difference of travel time STD and the difference of density heterogeneity of the same time interval pair. The results are shown in Fig. 3.

Based on Fig. 3, we observe that density heterogeneity's influence on travel time standard deviation (STD) follows a similar trend for both the shorter freeway section, I-694, and the longer one, I-805. The fluctuation in the ratio consistently demonstrates a left-skewed pattern relative to the average density. Specifically for the I-694 stretch, a peak effect on the travel time STD, equivalent to 0.79 min, is achieved at an average density of 17.5 vehicles per km per lane due to the growth in unit density heterogeneity. Meanwhile, for the longer and more congested I-805 stretch, the effects are amplified, peaking at 1.00 min when the average density reaches 32.8 vehicles per km per lane.

This phenomenon can be attributed to the state of freeway flow. When traffic is freeflowing, indicating a lower average density, an increase in density heterogeneity doesn't significantly trigger congestion or escalate travel time since only a few segments become congested. Similarly, when freeway density approaches its maximum, the impact of increased density heterogeneity on travel time reliability (TTR) is minimal, as many segments have already transitioned from a freeflow to a congested state.

However, the situation changes when density is close to congestion onset, around 7:00 am, or the end of offset, roughly 9:00 am. An uptick in density heterogeneity during these times can cause more segments to transition from freeflowing to congestion, significantly impacting the TTR.

In extreme scenarios where the freeway stretch is either devoid of vehicles or has reached its theoretical maximum density, the density heterogeneity stands at zero and does not affect travel time. Consequently, density heterogeneity exerts a more notable effect

Table 1
Notation.

Notation	Definition
A	arrival rate
$\overline{A}(t)$	average traffic flow of all segments in freeway stretch at time t
A_g	arrival rate for each lane, indexed by g
D	departure rate, constant and reflecting capacity
$E(n)$	expected number of vehicles
$E(T)$	expected waiting time
$E(T_J)$, $E(T_j)$	expected waiting time of the freeway stretch, segment j
$E(\mu_d)$	expected vehicle density for each lane
G	number of lanes in freeway stretch
J	number of segments in freeway stretch
L or l_j	length of freeway stretch or segment j
$r_{T_{j_1},T_{j_2}}$	Pearson correlation coefficient between waiting time of segments j_1 and j_2
$\overline{r}$	average Pearson correlation coefficients of travel times between all adjacent segments
T	waiting time, including service time and queueing time
T_s	waiting time threshold
α	parameter used to adjust the linear relationship between travel time mean and STD
β_j	ratio of the average arrival rate and the arrival rate of segment j
$\overline{\mu_d}$	average vehicle density of freeway stretch
μ_{d_j}	vehicle density of segment j in freeway stretch
μ_{dts}	vehicle density threshold
ρ	utilization rate
σ_{d_j}	difference between the vehicle density in segment j and the average vehicle density
σ_d	vehicle density heterogeneity of freeway stretch
σ_T or σ_{T_J}	waiting time STD or waiting time STD of the freeway stretch, which includes J segments
σ_τ^2	experienced travel time variance (TTV)
ζ_1, ζ_2, and ζ_0	parameters used to indicate the strength of the influence of average density, density heterogeneity and other factors on TTR

on the TTR near the onset or end of congestion compared to freeflowing or heavily congested conditions. This finding aligns with the effect of density heterogeneity on capacity as reported in Ramezani et al. (2015). Since most morning peak freeway travel typically occurs between congestion onset and offset, it is crucial to consider the impact of density heterogeneity on TTR. In the next section, we model the TTR of the freeway stretch using queueing theory.

3. Methodology

Queueing theory, a method utilized to examine the stochastic accumulation and scattering of systems as well as the operational processes of random service systems (Newell, 2013), is frequently employed in transport modeling (Heidemann, 1996; Vandaele et al., 2000; Han et al., 2013b; Cheng et al., 2022; Gao et al., 2026). We will next construct a theoretical model for TTR based on queueing theory, building upon the empirical findings from the previous section.

3.1. A brief review of queueing theory

To offer a benchmark for our forthcoming model, we initially apply queueing theory to investigate the relationship between vehicle density, density heterogeneity, and travel time STD prior to formulating a model using real data from our chosen freeway stretch. Key notations used in the model construction can be found in Table 1.

The M/M/1 model is a single-server queueing model without a capacity limit (Newell, 2013). Notably, 'capacity' in queueing models refers to the maximum number of vehicles that can concurrently occupy the freeway stretch. In reality, this state is seldom reached; therefore, using the uncapacitated model in this context is justified. In addition, the queues cannot expand without a limit on the freeway, which means that over the entire peak period, the queues are in an unsaturated state and remain stable, even if they are saturated for periods of time within the peak. Note that unsaturated here means that the queue length and waiting time will not go to infinity. There will still be congestion in the system because in the M/M/1 model, as long as the arrival rate A is greater than 0, there will be queueing time, that is, congestion. This model is based on several fundamental assumptions, as follows:

- The model assumes arriving vehicles follow a homogeneous Poisson process, signifying that the number of vehicles arriving within any given time period adheres to an independent Poisson distribution, and the time interval between arrivals obeys an exponential distribution, with an arrival rate of A.
- The service time, an independent random variable, adheres to an exponential distribution, reliant on the departure rate D, which can be conceptualized as the travel time in the absence of queueing.
- The distribution of queueing time is dictated by the arrival and departure rates A and D, and represents the additional time vehicles spend on the freeway stretch beyond the service time.
- The waiting time of vehicles on the freeway stretch adheres to the exponential distribution of parameters $D - A$.

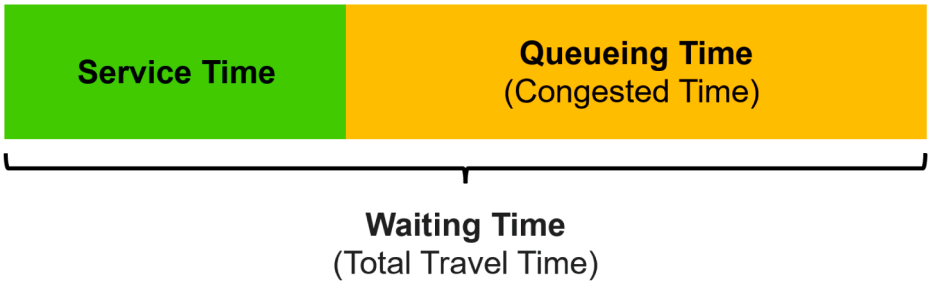


Fig. 4. Schematic diagram of travel time based on M/M/1 model.

A higher arrival rate A and lower departure rate D translate into increased service and queueing time, signifying prolonged wait times for vehicles on the freeway stretch. The waiting time T is the sum of the service and queueing time, representing the total time a vehicle spends traversing the freeway stretch, as shown in Fig. 4.

Day-to-day travel time variability on the freeway stretch primarily results from fluctuations in daily traffic demand and changes in capacity. Although numerous prior studies have considered vehicles on the freeway as Poisson flow (Yuan et al., 2022; Yue et al., 2016; Willsky et al., 1980), it's crucial to confirm that vehicle arrivals on the freeway stretch adhere to the Poisson process before employing the queueing model to simulate the TTR. This confirmation is achieved by determining whether vehicle arrival time intervals conform to an exponential distribution at four time points during the morning peak period. Additionally, it's important to note that freeway stretch ramps, particularly on-ramps, often cause recurring congestion due to merging traffic, thereby creating significant long-term bottlenecks (Zhang and Levinson, 2010). We evaluate the probability distribution characteristics of the arrival rate for segments with on-ramps and without on-ramps separately, and have:

Proposition 1. *Regardless of the presence of ramps, vehicle arrivals on the freeway conform to Poisson flow characteristics, meaning the arrival intervals follow an exponential distribution with the parameter being the arrival rate A.*

The detailed discussion is shown in Appendix C. It should be noted that the M/M/1 model referred to here is not based on vehicle-to-vehicle interval distribution, as we are aiming to simulate day-to-day fluctuations in travel time. Therefore, the corresponding interval distribution is also day-to-day. However, this does not affect the mathematical properties of the model itself. In addition, it's observed that the arrival rate A fluctuates considerably at different time points during the morning peak period, indicating a time in-homogeneous Poisson process for different departure time intervals. Therefore, the arrival rate A in the model should be a dynamic variable.

Further, we note that traffic flow and arrival time intervals have a reciprocal relationship, leading to day-to-day distributions that are symmetric and skew-dependent. Daily average results for traffic flow and Pearson correlation coefficients between traffic flow and the estimated arrival rate are presented in Table C.1. These results demonstrate that traffic flow effectively measures the arrival rate's strength for both segments with and without on-ramps. Given the strong autocorrelation and day-to-day pattern consistency on workdays within morning peak traffic flow, it's concluded that vehicle arrivals on a freeway stretch can be modeled as a homogeneous Poisson process with variable arrival rate A. This signifies that, within short time intervals, vehicle arrival intervals adhere to an exponentially independent and identically distributed (IID) principle. In the following model, we use the daily average traffic flow at each departure time interval to dynamically represent the arrival rate A. It should be noted that, since the arrival rate A is updated using real-time observed data, the spatial continuity relationship in deterministic queueing theory (Gao and Levinson, 2024b; Cheng et al., 2022) is naturally reflected in the stochastic queueing formulation.

In addition, for the M/M/1 model, an important characteristic is the independence of the distribution of vehicle arrival intervals and service times. To further illustrate the validity of using the M/M/1 model, we confirm that the distributions of vehicles' service time and the vehicle arrival intervals, i.e. vehicle arrival intensity, are mutually independent. The specific details regarding this are shown in Appendix D. In real-world scenarios, the service and queueing times of vehicles are intertwined and not distinctly separated in time and space. This means that the service and queueing of vehicles cannot be clearly separated and are inconsistent with the assumptions of the M/M/1 model. However, this simplifying assumption does not affect the practicality of the model when considering linear additivity of travel times, and is similar to previous research (Vickrey, 1969; Zhou et al., 2022; Han et al., 2013a) assumptions where the accumulated vehicles, i.e. queue, are only stacked at the downstream bottleneck, while the upstream sections are in freeflow state.

Therefore, after determining the independence of service time and vehicle arrival interval, we next verify whether the total travel time, i.e., the waiting time, adheres to the exponential distribution with parameter $D - A$ to ascertain if it fulfills another of M/M/1 model's fundamental properties. We have:

Proposition 2. *The total travel time of vehicles on the freeway follows an exponential distribution with the parameter $D - A$, where the departure rate D remains stable over time, regardless of the presence of ramps.*

The specific details are illustrated in Appendix E. It should be noted that the departure rate D in the M/M/1 model does not refer to the freeway's capacity to process traffic flow, but rather indicates the distribution of service time. The larger the departure rate D, the

more right-skewed the distribution of service time. From Table E.1, We observe that the departure rate D for non-busy periods during the morning peak is slightly higher than that during busy times due to capacity drops (Saberi and Mahmassani, 2012; Geroliminis and Sun, 2011a). This will affect the parameters of the service time distribution, but it does not affect the fundamental assumption of independence between vehicle arrival and service time in the M/M/1 model. This also indicates that the freeway capacity can be used to indicate the departure rate D.

Consequently, Propositions 1 and 2 empirically support the basic M/M/1 conditions of Poisson arrivals and exponentially distributed travel times. We therefore approximate each freeway segment as a single-server system and use the uncapacitated M/M/1 model as a parsimonious and analytically tractable representation. This choice improves scalability because multiple freeway segments can be connected in a tractable multi-segment framework, while its adaptability depends on whether the arrival and travel time distributions in new contexts remain close to the M/M/1 assumptions. We directly use the outputs of the M/M/1 model, bypassing the derivation process here. These outputs include the expected waiting time $E(T)$ for incoming vehicles, which can be perceived as experienced travel time; the probability $P(T \le T_s)$ of a vehicle spending time T_s or less in the system; and the expected number of vehicles $E(n)$ in the system, as shown in Eqs. (1)–(3).

$$E(T) = \frac{1}{D - A} \tag{1}$$

$$P(T \le T_s) = 1 - e^{-(1-\rho)DT_s} \tag{2}$$

$$E(n) = \frac{A}{D - A} \tag{3}$$

In these equations, A represents the arrival rate, which is a dynamic variable, namely $A(t)$, and D represents the departure (or service) rate, which is treated as a constant. $\rho = \frac{A}{D}$ refers to the utilization rate, and ρ must be less than 1 to maintain model stability and to ensure that the queue in the system does not extend infinitely. A noteworthy point is that $E(n)$ and $E(T)$ adhere to Little's Formula (Little, 1961), which is given by $E(n) = A \cdot E(T)$. Building on the cumulative distribution function (CDF) in Eq. (2), we can express the probability density function (PDF) of waiting time, i.e. total travel time, T in the freeway stretch as presented in Eq. (4).

$$\begin{aligned} P(T = T_s) &= \frac{\mathrm{d}P(T \le T_s)}{\mathrm{d}T} \\ &= (D - A) \cdot e^{-(D-A)T_s} \end{aligned} \tag{4}$$

3.2. Measuring travel time reliability based on queueing theory

Based on these outcomes derived from the M/M/1 model, we assess the standard deviation of travel time. The term under the square root in Eq. (5) represents variance, which, by definition, quantifies the extent of deviation between a random variable and its expected value. The standard deviation of total travel time σ_T (Servi and Finn, 2002) for vehicles passing through a specific freeway stretch segment is provided in Eq. (5).

$$\begin{cases} \sigma_T = \sqrt{E\left((T - E(T))^2\right)} \\ \quad = \sqrt{E(T^2) - E(T)^2} \\ \quad = \sqrt{\displaystyle\int_{T=0}^{+\infty} (D - A) \cdot e^{-(D-A)T} \cdot T^2 \,\mathrm{d}T - \left(\frac{1}{D - A}\right)^2} \\ \quad = \dfrac{1}{D - A} = E(T) = \dfrac{E(n)}{A} \\ \sigma_T = \dfrac{E(n)}{A} = \dfrac{l \cdot G \cdot E(\mu_d)}{A} = \dfrac{l \cdot G \cdot E(\mu_d)}{A_g \cdot G} = \dfrac{l \cdot E(\mu_d)}{A_g} = \dfrac{l}{E(V)} = E(T) \end{cases} \tag{5}$$

Where we set the length and the number of lanes of the freeway stretch to be L and G respectively, the arrival rate for each lane is A_g, and the length of each segment is equal to l. Therefore, the vehicle density for each lane μ_d of a certain segment on the freeway stretch is equal to $\frac{E(n)}{l \cdot G}$.

According to Eq. (5), based on the basic properties of the M/M/1 queueing model, we know the waiting time STD, i.e. total travel time STD, σ_T is equal to expectation $E(T)$. Similarly, the second equation in Eq. (5) also gives a self-consistent result, however, this is based on the basic relationship between flow, density, and speed. This derivation demonstrates the viability of the M/M/1 model to describe the relationship between travel time reliability and vehicle density. The linear relationship between travel time and travel time STD derived from the model is consistent with existing theories (Torrisi et al., 2017; Mahmassani et al., 2012; Saedi et al., 2020), which show travel time variability of day-to-day, vehicle-to-vehicle, and link-to-link has a nearly linear relationship with mean travel time based on empirical studies with the normalized length of the segments. Therefore, the M/M/1 queueing model also provides hints for explaining various types of travel time variability besides day-to-day.

In addition, according to Eq. (5), we can also provide a theoretical explanation for the hysteresis loops in the relationship between average density and experienced travel time STD in Fig. 2(a). Under the same density $E(\mu_d)$, a lower arrival rate A_g will lead to a

higher travel time STD. Based on previous empirical research (Zhang and Levinson, 2010; Geroliminis and Sun, 2011a; Saberi and Mahmassani, 2012), we know that the freeway throughput during the congestion offset period will be lower than the onset period due to lower capacity and arrival flow. This reduced throughput results from higher spatial density heterogeneity, which we will discuss in detail in subsequent sections. Therefore, under the premise of satisfying the basic properties of the M/M/1 queueing model, the travel time fluctuations in the congestion offset period will be more severe compared to the onset period.

However, it is difficult for this simple model to explain the empirical hysteresis loops obtained from the travel time mean and STD verified by Kim and Mahmassani (2015), Yildirimoglu et al. (2015), because the same mean travel time means the same travel time STD based on the model. There are several reasons as shown below.

- The M/M/1 model assumes that the vehicle arrival satisfies the Poisson process, and according to the results of Fig. C.1, the reliability of this assumption is relatively weak at the beginning and end of the morning peak period.
- The waiting time expectation $E(T)$ and STD σ_T are considered t integrals range from 0 to infinity, while vehicle travel time t is constrained and surely less than infinity.
- Finally, the traffic flow and vehicle density differ for different segments on the freeway stretch during the morning peak period, and the model does not consider this heterogeneity.

Even so, the results of the M/M/1 model can well validate our empirical common sense, that is, the experienced travel time STD is inversely proportional to the driving speed, without the need for more freeway condition data. Next, we will focus on analyzing the impact of the density heterogeneity of segments on TTR.

3.3. Accounting for covariance of travel time between segments

We presume that the freeway stretch comprises a number of segments denoted as J, each of length l and having the same number of lanes G. These J segments can be interpreted as J interconnected M/M/1 single servers. This is based on the results of our earlier analysis which established that regardless of the presence of an on-ramp, the standard deviation of waiting time can be described by the M/M/1 model during the morning peak period. However, before deducing the distribution of the waiting time standard deviation for the entire freeway stretch, we need to ascertain whether the waiting times for these J segments are independent. Intuitively, they are interrelated, particularly the adjacent segments, which exhibit strong positive correlations (Rakha et al., 2006; Sumalee et al., 2006; Ramezani and Geroliminis, 2012) and contribute to the increase in TTV for the entire freeway stretch.

We cannot consider the travel times of different segment pairs to be independent of one another (Correlation matrices are discussed in Appendix F), so the standard deviation of waiting time σ_{T_J} for the entire freeway stretch is expressed in Eq. (6).

$$\begin{aligned}\sigma_{T_J} &= \sqrt{\sum_{j=1}^{J}(\sigma_{T_j})^2 + \sum_{1\le j_1,j_2\le J} Cov(T_{j_1},T_{j_2})} \\ &= \sqrt{\sum_{j=1}^{J}(\sigma_{T_j})^2 + \sum_{1\le j_1,j_2\le J} r_{T_{j_1},T_{j_2}}\cdot\sigma_{T_{j_1}}\cdot\sigma_{T_{j_2}}}\end{aligned} \tag{6}$$

In the formula, σ_{T_j} represents the standard deviation of waiting time for segment j, $Cov(T_{j_1},T_{j_2})$ represents the covariance of waiting times between segments j_1 and j_2, and $r_{T_{j_1},T_{j_2}}$ is the Pearson correlation coefficient for waiting times between segments j_1 and j_2. Combining Eq. (6) with Eq. (5), we have Eq. (7).

$$\begin{aligned}\sigma_{T_J} &= \sqrt{\sum_{j=1}^{J}(\alpha\cdot E(T_j))^2 + \sum_{1\le j_1,j_2\le J} r_{T_{j_1},T_{j_2}}\cdot\alpha\cdot E(T_{j_1})\cdot\alpha\cdot E(T_{j_2})} \\ &= \alpha\cdot\sqrt{\sum_{j=1}^{J}(E(T_j))^2 + \sum_{1\le j_1,j_2\le J} r_{T_{j_1},T_{j_2}}\cdot E(T_{j_1})\cdot E(T_{j_2})}\end{aligned} \tag{7}$$

Where $E(T_j)$ is the expected travel time of segment j, and α is the parameter used to adjust the linear relationship between travel time mean and STD for each segment, which is generally less than 1 (Mahmassani et al., 2012). Based on Eq. (7), for the parameter α of the mean and STD of travel times for different distances, we have:

Proposition 3. *For a homogeneous segment, as the travel distance increases, the parameter α between the travel time mean and STD will decrease.*

Proof For any freeway segment j with a length of l, we consider it as being composed of two sub-segments j_1^* and j_2^*, each with a length of $\frac{l}{2}$. Since it satisfies homogeneity, the relationship between the travel time mean and STD for each sub-segment is:

$$\sigma_{T_{j^*}} = \alpha^*\cdot E(T_{j^*}) \tag{8}$$

Where α^* is a constant used to describe the linear relationship between the mean and the STD of travel time for each sub-segment. Therefore, based on Eq. (7), for segment j, the relationship between its travel time mean and STD is given by:

$$\sigma_{T_j} = \alpha^* \cdot \sqrt{E(T_{j_1^*})^2 + E(T_{j_2^*})^2 + 2 \cdot r_{T_{j_1^*}, T_{j_2^*}} \cdot E(T_{j_1^*}) \cdot E(T_{j_2^*})} \tag{9}$$

Where $r_{T_{j_1^*}, T_{j_2^*}}$ is the Pearson correlation coefficient of travel time between sub-segments j_1^* and j_2^*. In addition, for segment j, we know that the relationship between its travel time mean and STD is:

$$\begin{aligned} \sigma_{T_j} &= \alpha \cdot E(T_j) \\ &= \alpha \cdot (E(T_{j_1^*}) + E(T_{j_2^*})) \\ &= \alpha \cdot \sqrt{E(T_{j_1^*})^2 + E(T_{j_2^*})^2 + 2 \cdot E(T_{j_1^*}) \cdot E(T_{j_2^*})} \end{aligned} \tag{10}$$

From Eqs. (9) and (10), we observe that $2 \cdot r_{T_{j_1^*}, T_{j_2^*}} \cdot E(T_{j_1^*}) \cdot E(T_{j_2^*}) \leq 2 \cdot E(T_{j_1^*}) \cdot E(T_{j_2^*})$. In practical scenarios, the equality here is typically not attainable. Therefore, $\alpha^* > \alpha$. Consequently, by iterating this method, it can be shown that as travel distance increases, the parameter α decreases. The proof is completed.

Proposition 3 illustrates that as the travel distance increases, the magnitude of travel time fluctuations weakens relative to the average travel time. For instance, a trip that usually takes 5 min might increase to 10 min due to congestion, representing a 100% fluctuation. However, if the travel time is 1 h, the congestion is unlikely to double it to 2 h, as some segments may be congested while others remain free flow. This suggests that the travel time correlations between different segments do not follow a perfectly positive linear relationship, which aligns with real-world experience. This also implies that the strength of travel time correlations between different segments affects the magnitude of overall travel time fluctuations. This suggests that, for long freeway stretches, reliability-oriented traffic management should not only address individual bottlenecks, but also reduce the correlation of travel time fluctuations across segments by preventing bottleneck spillback and congestion propagation through coordinated control measures such as ramp metering and variable speed limits.

As per the Pearson correlation coefficient matrices displayed in Figs. F.1 and F.2, the average Pearson correlation coefficients for different segment pairs on freeway stretch I-694 are observed to be 0.602, 0.648, 0.742, and 0.718 at times 6:30, 7:30, 8:30, and 9:30, respectively. Similarly, for the freeway stretch I-805, these values are found to be 0.363, 0.457, 0.608, and 0.514. This means that the travel time correlations between different segments during the congestion offset are significantly higher than the congestion onset. And for the I-805 with multiple bottlenecks, this gap is even more obvious.

Therefore, for the empirical hysteresis loops in travel time mean and STD that Eq. (5) cannot explain, we can provide an explanation under the consideration of the covariance of travel times across different segments. We have:

Proposition 4. *Under the same total travel time, the travel time STD during the congestion offset period is higher than that during the onset period, due to a higher spatial density heterogeneity during the congestion offset period and a stronger correlation in travel times across different segments.*

Proof We know that under the same expected total travel time $E(T_J)$, the travel time of different segments $E(T_j)$ corresponding to two different time points in the congestion onset and offset period should also be roughly equal, since queue formation and dissipation are roughly symmetric in time and space (Cheng et al., 2022). However, due to the higher density heterogeneity during the congestion offset period as shown in Fig. 2(b), this results in more significant differences in travel times among different segments under the same total travel time. For the entire freeway stretch, the expected total travel time $E(T_J)$ is equal to the sum of the expected travel times of each segment, even if they are not independent of each other, as shown in Eq. (11).

$$\begin{aligned} E(T_J) &= E(\sum_{j=1}^{J} T_j) \\ &= \sum_{j=1}^{J} E(T_j) \end{aligned} \tag{11}$$

Therefore, as for the first term $\sum_{j=1}^{J}(E(T_j))^2$ in Eq. (7), we know that under the condition of constant total travel time, i.e., $\sum_{j=1}^{J} E(T_j)$ as shown in Eq. (11) equals a constant, the equal travel times for all segments will lead to the minimum value of $\sum_{j=1}^{J}(E(T_j))^2$. Specific proof details are shown in Appendix G. Therefore, during the congestion offset period, the higher density heterogeneity causes the first term in Eq. (7) of total travel time STD to have larger values. Additionally, the second term in Eq. (7) experiences greater correlations among the travel times of different segment pairs during the congestion offset period, which will also cause higher total travel time STD due to higher covariance, under the condition that the travel times of different segments are approximately equal during both the congestion onset and offset periods. Thus, overall, we know that under the same total travel time, the travel time STD in the congestion offset period is higher than the onset period. The proof is completed.

Then, we compare the results of mean travel time and travel time STD of the selected freeway stretches with the results derived from the model Eq. (7). Here, we consider covariance relationships of different orders, from zero order covariance to full order covariance to test the impact of covariance on model estimation performance as shown in Fig. 5.

The order represents the farthest distance from the current segment to other segments. For example, zero order covariance means only considering the standard deviation of each segment, which is the black square in Fig. 5. First order covariance means considering

Table 2
TTR modeling considering covariances of different orders (Eq. (7)).

	I-694			I-805		
	α	RMSE	R^2	α	RMSE	R^2
No covariance		0.713	0.175		3.709	0.192
First order covariance		0.321	0.634		1.563	0.512
Third (Fifth) order covariance	0.45	0.162	0.865	0.41	0.755	0.811
Fifth (Ninth) order covariance		0.121	0.925		0.450	0.933
Full covariance		0.117	0.928		0.440	0.936

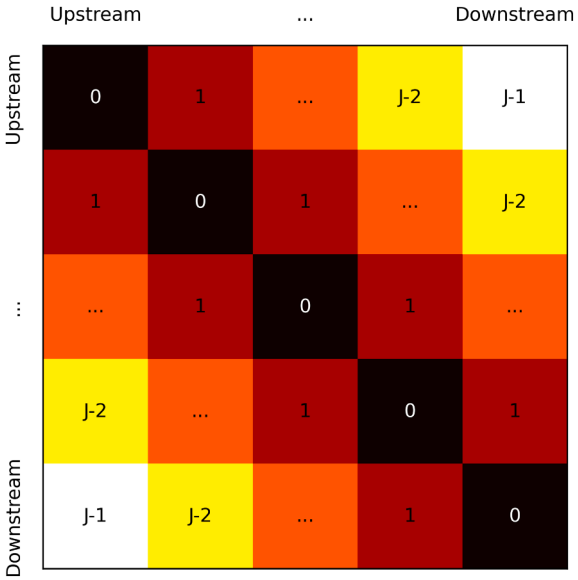
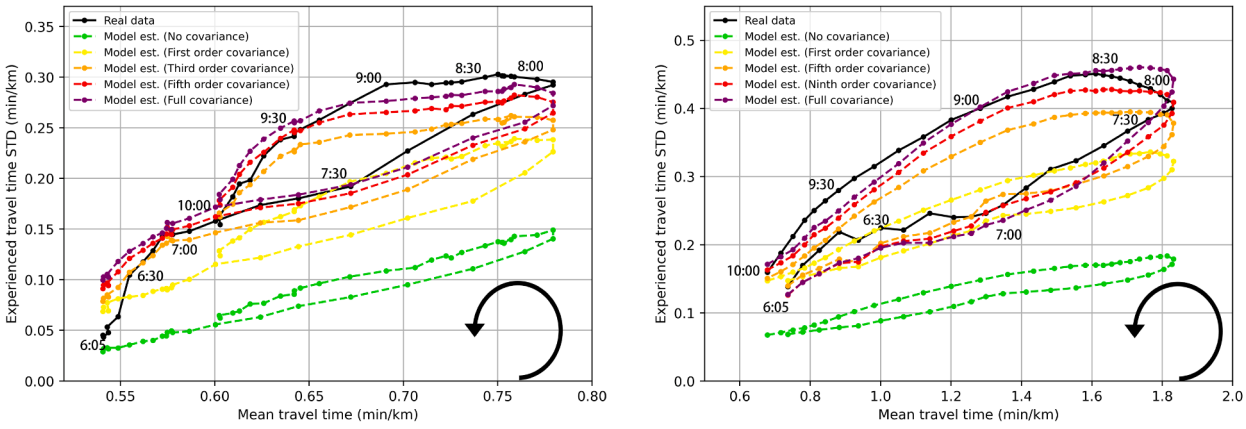


Fig. 5. Schematic diagram of covariance of different orders.



(a) I-694 (b) I-805

Fig. 6. Comparison between the model estimation and the real data of the experienced travel time STD.

the covariance between no more than one adjacent segment of the current segment in addition to the respective standard deviations, i.e. the black squares and red squares in the figure. Full order covariance means considering all $J-1$ adjacent segments covariance relationship, that is, all squares. The estimate of parameter α is the average of the estimated results of the linear relationship between the travel time mean and STD of each segment since segments on the same freeway stretch have similar characteristics. The results are shown in Table 2 and Fig. 6.

From Table 2 and Fig. 6, we observe that when the covariance between segments is not considered, the counterclockwise hysteresis loops can still be reproduced but are mislocated, because the sum of variances of all segments also differs in congestion onset and

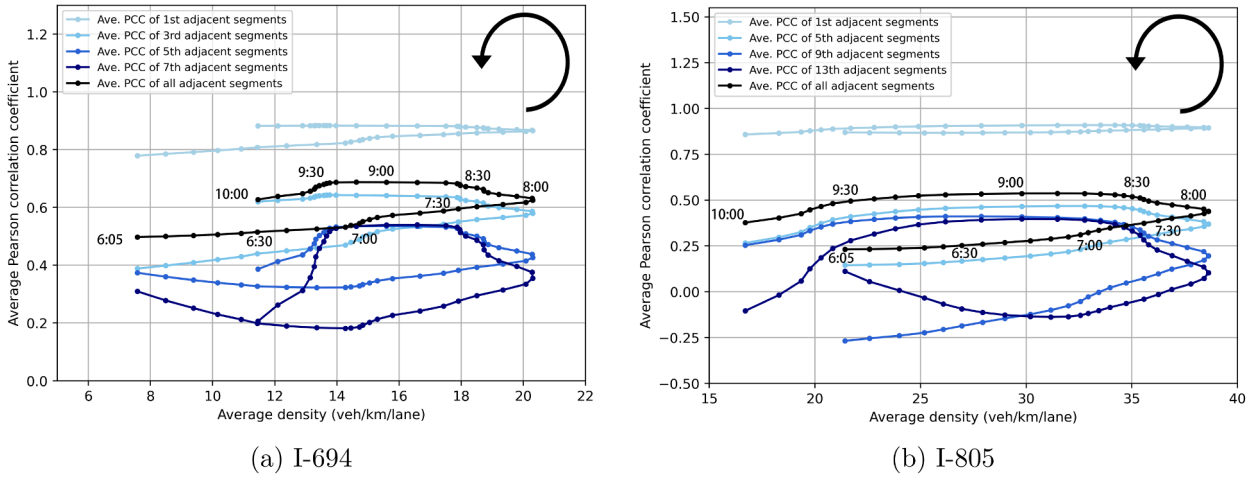


Fig. 7. Curves of average vehicle density and average Pearson correlation coefficient in different adjacent distances.

offset periods. However, after considering covariance, the performance of the model is significantly improved because the covariance caused by strong correlation contributes significantly to the overall variance and the travel time correlation of different segment pairs is stronger in the congestion offset period. Therefore, the test results are consistent with our discussion in the proof of Proposition 4. This also verifies the previous empirical research on the hysteresis loops between the travel time mean and variance (Yildirimoglu et al., 2015).

As to why the travel times of different segment pairs have a higher correlation during the congestion offset period. We speculate that this is due to higher density heterogeneity in the offset period. Next, we plot the curves of the average density and average Pearson correlation coefficient in different adjacent distances of the freeway stretches as displayed in Fig. 7.

We observe that during the congestion offset period, the Pearson correlation coefficient of the travel time of different segment pairs is higher than that during the congestion onset period, and this applies to segment pairs with different adjacent distances. Combining Fig. 7 and Fig. 2(b), it becomes evident that with the same average density, greater density heterogeneity correlates with a higher average Pearson correlation coefficient. Furthermore, the extent of variation in density heterogeneity aligns with the extent of variation in the average Pearson correlation coefficient.

We know that on a freeway stretch, different segment pairs exhibit higher correlations in travel times when experiencing either freeflow traffic or congestion. This is because their travel times tend to be either close to freeflow speeds or significantly delayed due to congestion, resulting in relatively minor differences between them. Hence, with a rising average density, the correlation in travel times among various segments increases owing to the escalating congestion. This is reflected in the changes in the Pearson correlation coefficient from 6:30 to 7:30, and from 8:30 to 9:30 in Figs. F.1 and F.2, which experience an increase or decrease in average density.

However, under the same average traffic density conditions, an increase in density heterogeneity causes all segments within the freeway stretch to be more likely to experience either non-congestion or more extreme states of congestion. This leads to higher travel time correlations within both congested and non-congested segments individually.

And the correlation between these two categories of segments will also strengthen. This is because the links in each of these two categories of segments tend to be spatially adjacent (congestion clusters, the absence of congestion clusters), while the typical link in each category is spatially separated from the typical link in the other category (a congested link is farther from a non-congested link). Higher density heterogeneity will cause congestion segments located in bottlenecks generating more severe backward shockwaves, thereby causing an increase in travel time for previously non-congested segments as well. This is reflected in the changes in the Pearson correlation coefficient from 6:30 to 9:30, and from 7:30 to 8:30 in Figs. F.1 and F.2, which experience increased density heterogeneity under roughly the same average density.

Therefore, overall, regarding the higher time correlation between segments during congestion offset as stated in Proposition 4, it is also due to higher density heterogeneity leading to increased travel time correlations within and between different categories of segments, consequently resulting in higher total travel time STD during congestion offset periods.

3.4. Accounting for only first order covariance

While the model considering full order covariance, as per Eq. (7), can offer a reliable estimate of the total travel time STD, it has its limitations in practical application. It relies on having access to the travel times of all individual segments and their corresponding Pearson correlation coefficients. Moreover, as the number of segments increases, the number of coefficients also grows quadratically, i.e. $O(n^2)$, imposing significant constraints on the model's practical applicability.

According to the test results shown in Table 2 and Fig. 6, after considering the first order covariance, the model performance significantly improves because adjacent segments have a stronger travel time correlation compared to other more distant segments.

Therefore, it is reasonable to test a model that only accounts for the covariance with adjacent segments, which can strike a balance between interpretability and complexity. However, due to the loss of partial covariance, the model needs to be further calibrated.

The standard deviation of waiting time σ_{T_J} for the entire freeway stretch based on Eq. (6) simplifies to Eq. (12).

$$\begin{aligned}\sigma_{T_J} &\approx \sqrt{\sum_{j=1}^{J}(\sigma_{T_j})^2 + \sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}} Cov(T_{j_1},T_{j_2})} \\ &= \sqrt{\sum_{j=1}^{J}(\sigma_{T_j})^2 + \sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}} r_{T_{j_1},T_{j_2}}\cdot\sigma_{T_{j_1}}\cdot\sigma_{T_{j_2}}}\end{aligned} \tag{12}$$

As per the Pearson correlation coefficient matrices displayed in Figs. F.1 and F.2, the Pearson correlation coefficients for travel times between all adjacent segments appear similar, so using the average value, denoted by $\bar{r}$, simplifies the model. As for the first order Pearson correlation coefficients, their temporal variation is slight, so $\bar{r}$ can be treated as a constant.

While this prevents us from using dynamic Pearson correlation coefficients to capture differences in travel time STD during different congestion periods, we instead achieve this through the dynamic inputs in traffic flow. In reality, using Pearson correlation coefficients as a predictable dynamic variable input into the model is challenging. The higher total travel time STD caused by the sum of variances of each segment and the sum of their covariances in Eq. (7) results from the increase in density heterogeneity. Therefore, the constant $\bar{r}$ does not impact the overall interpretability of the model since it only affects the covariance term. By combining Eq. (5) with Eq. (12), we obtain Eq. (13), which presents the standard deviation of waiting time σ_{T_j} for the freeway stretch.

$$\begin{aligned}\sigma_{T_J} &\approx \sqrt{\sum_{j=1}^{J}\left(\frac{\mu_{d_j}\cdot l\cdot G}{A_j}\right)^2 + \bar{r}\cdot\sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}} \frac{\mu_{d_{j_1}}\cdot l\cdot G}{A_{j_1}}\cdot\frac{\mu_{d_{j_2}}\cdot l\cdot G}{A_{j_2}}} \\ &= \sqrt{\sum_{j=1}^{J}\left(\frac{\mu_{d_j}\cdot l\cdot G\cdot\beta_j}{\overline{A}}\right)^2 + \bar{r}\cdot\sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}} \frac{\mu_{d_{j_1}}\cdot l\cdot G\cdot\beta_{j_1}}{\overline{A}}\cdot\frac{\mu_{d_{j_2}}\cdot l\cdot G\cdot\beta_{j_2}}{\overline{A}}}\end{aligned} \tag{13}$$

Where $\overline{A}$ is the average arrival rate, i.e. average traffic flow, of all segments on the freeway stretch, A_j is the arrival rate for segment j, and β_j is the ratio of the average arrival rate and the arrival rate of segment j, which is equal to $\frac{\overline{A}}{A_j}$. Before we regard $\{\beta_1,\beta_2\dots\beta_J\}$ as parameters, we need to ensure that it is stable in time series. We have:

Proposition 5. *The relative magnitude of density fluctuates more significantly than traffic flow across different segments on a freeway stretch.*

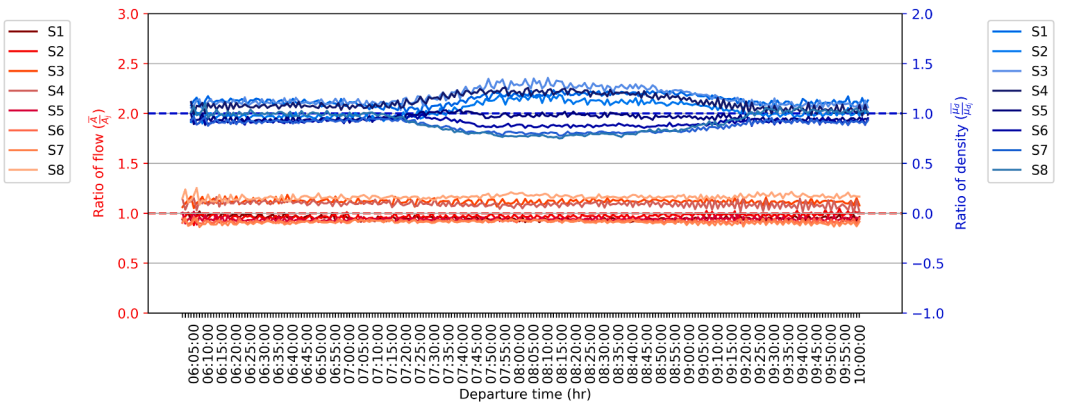
Remark 5 The change in the ratio of the average arrival rate of the freeway stretch to the arrival rate for each of the segments on I-694 and I-805 during the morning peak period are shown in Fig. 8(a) and (b), respectively. In addition, we also provide the changes of the ratio $\frac{\overline{\mu_d}}{\mu_{d_j}}$ of the average density and the density of each segment in time for comparison.

As per Fig. 8, it can be observed that the set $\{\beta_1,\beta_2\dots\beta_J\}$ displays minor fluctuations around certain values, from 6:00 am to 10:00 am. For the single bottleneck freeway stretch I-694, it is more stable in time. This indicates that the relative intensity of traffic flows across different segments remains consistent during the morning peak hours. This consistency is largely due to commuting being the primary activity during this period, resulting in relatively stable spatial travel distribution (Geroliminis and Sun, 2011a). For the freeway stretch I-805 with multiple bottlenecks, although the traffic flow ratio $\frac{\overline{A}}{A_j}$ fluctuates over time due to the combined effect of multiple shockwaves, it appears negligible compared to the fluctuations in density ratio $\frac{\overline{\mu_d}}{\mu_{d_j}}$. This is because the temporal variations in density are essentially the accumulation of spatial differences in traffic flow over time (Daganzo, 1995) as shown in Eq. (14).

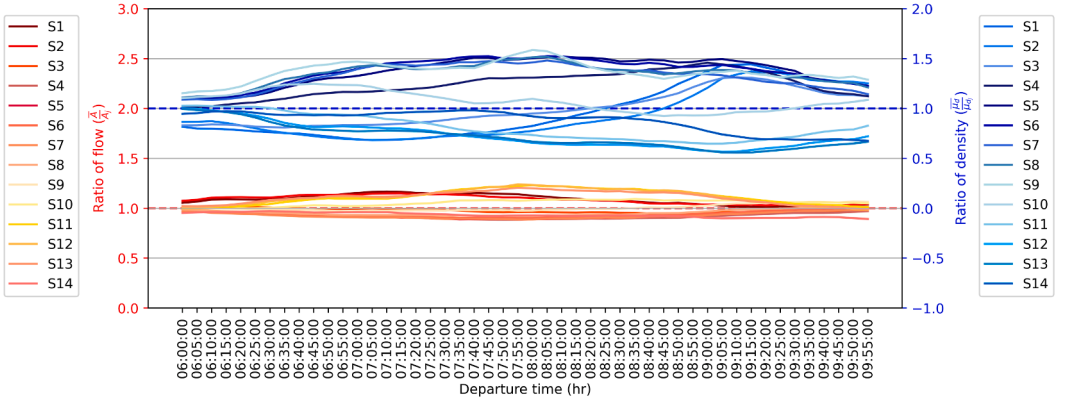
$$\frac{\partial\mu}{\partial t}+\frac{\partial A}{\partial x}=0 \tag{14}$$

According to the LWR equation (Lighthill and Whitham, 1955; Richards, 1956) shown in Eq. (14), we can understand that even if the flow remains relatively stable over time, the density will still accumulate due to flow difference in space, leading to more significant temporal fluctuations. Thus, the set $\{\beta_1,\beta_2\dots\beta_J\}$ can be considered as parameters, and the dynamic variations in congestion among different segments will be captured through density.

Furthermore, the difference between the arrival rates for most segments and the average arrival rate, $\overline{A}$, falls within the $\pm10\%$ range. This limited deviation also serves as an empirical robustness check for the approximation of segment-level arrival rates. This suggests that there are no severe bottlenecks, and the flow on the freeway stretch is smooth. To simplify the subsequent formula derivation and obtain a practical expression about TTR, all these parameters are taken as 1. The loss of variation in arrival rates among different segments caused by this will be manifested by the density heterogeneity of the segments. However, it is crucial to include calibration parameters in the final form to compensate for any resultant error.



(a) I-694



(b) I-805

Fig. 8. Change in the ratio of the average arrival rate and average density of the freeway stretch to the arrival rate and density for each of the segments.

$$
\begin{aligned}
\sigma_{T_J} &\approx \sqrt{\sum_{j=1}^{J}\left(\frac{\mu_{d_j}\cdot l\cdot G}{\overline{A}}\right)^2+\overline{r}\cdot\sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}}\frac{\mu_{d_{j_1}}\cdot l\cdot G}{\overline{A}}\cdot\frac{\mu_{d_{j_2}}\cdot l\cdot G}{\overline{A}}} \\
&= \frac{l\cdot G}{\overline{A}}\sqrt{\sum_{j=1}^{J}\left(\overline{\mu_d}+\sigma_{d_j}\right)^2+\overline{r}\cdot\sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}}\left(\overline{\mu_d}+\sigma_{d_{j_1}}\right)\cdot\left(\overline{\mu_d}+\sigma_{d_{j_2}}\right)} \\
&= \frac{l\cdot G}{\overline{A}}\sqrt{\sum_{j=1}^{J}\left(\overline{\mu_d}+\sigma_{d_j}\right)^2+\overline{r}\cdot\sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}}\left(\overline{\mu_d}^2+(\sigma_{d_{j_1}}+\sigma_{d_{j_2}})\cdot\overline{\mu_d}+\sigma_{d_{j_1}}\cdot\sigma_{d_{j_2}}\right)} \\
&= \frac{l\cdot G}{\overline{A}}\sqrt{\left(J+2\overline{r}\cdot(J-1)\right)\cdot\overline{\mu_d}^2+(2+4\overline{r})\cdot\overline{\mu_d}\cdot\sum_{j=1}^{J}\sigma_{d_j}+\sum_{j=1}^{J}\sigma_{d_j}^2-\overline{r}\left(2\overline{\mu_d}\cdot(\sigma_{d_1}+\sigma_{d_J})-\sum_{\substack{1\le j_1,j_2\le J,\\ |j_1-j_2|=1}}\sigma_{d_{j_1}}\cdot\sigma_{d_{j_2}}\right)}
\end{aligned}
\tag{15}
$$

Where μ_{d_j} is the vehicle density in segment j, and σ_{d_j} is the difference between the vehicle density in segment j and the average vehicle density $\overline{\mu_d}$ in the freeway stretch. We know the term $\sum_{j=1}^{J}\sigma_{d_j}$ in Eq. (15) equals $\sum_{j=1}^{J}\left(\mu_{d_j}-\overline{\mu_d}\right)$, which means the sum of the deviations of the vehicle density of all segments from the average density and is equal 0. And in the last term of Eq. (15), the first

item represents the product of the vehicle density deviation of the first and last segments and the average vehicle density, and the second item is the summation of the product of the vehicle density deviations of any two adjacent segments. We determine whether these two items can be ignored as minor by obtaining the ratio of these two items to the other terms respectively. $Ratio^1$ and $Ratio^2$ represent the ratio of the first item and the second item of the last term to the other terms, which are shown in Eqs. (H.1) and (H.2), respectively, of Appendix H.

Based on the data for freeway stretches I-694 and I-805, the vehicle density across different segments varies by no more than 40% from the average vehicle density of the entire freeway stretch across all departure time intervals. Hence, by integrating Eqs. (H.1) and (H.2), we derive that the upper limits for $Ratio^1$ are 8.33% and 4.28% for freeway stretches I-694 and I-805 respectively. The upper limits for $Ratio^2$ are 9.50% and 2.76% for the same freeway stretches. Note that these upper limit values decrease as the number of segments increases. Considering that we don't account for the positive and negative conditions of the two ratios and have ignored certain items in the derivation process, the actual result is likely significantly lower than these upper limit values. Indeed, real data from these two freeway stretches corroborate this. For freeway stretch I-694, across different departure time intervals, $Ratio^1$ and $Ratio^2$ range from 0.01% to 0.86% and 0.05% to 0.57% respectively. For freeway stretch I-805, the $Ratio^1$ and $Ratio^2$ values span from 0.01% to 0.41% and 0.02% to 0.16% respectively. Thus, we can treat the final term of Eq. (15) as negligible and dismiss it.

During the derivation of Eq. (15), we did not add a parameter α to adjust the linear relationship between travel time STD and mean travel time (Mahmassani et al., 2012), and we disregard the minor term and approximate the parameters $\{\beta_1, \beta_2 \dots \beta_J\}$. In addition, we only considered the covariance of adjacent segments. Therefore, it's essential to include calibration parameters and constant error terms in the model, as presented in Eq. (16).

$$\sigma_{T_J} = \frac{l \cdot G}{\overline{A}} \sqrt{\zeta_1 \cdot \left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2 + \zeta_2 \cdot \sum_{j=1}^{J} \sigma_{d_j}^2 + \zeta_0} \tag{16}$$

Where ζ_1 and ζ_2 can be regarded as parameters to quantify the influence intensity of average density and density heterogeneity on TTR respectively. ζ_0 is an error term, whose unit is $(veh/km/lane)^2$ to ensure the unity of dimensions. It compensates for the impact of errors on TTR caused by neglect or approximation during the derivation process, and also includes other influencing factors such as weather, driving behavior, etc. The vehicle density heterogeneity σ_d refers to the degree of deviation of vehicle density from the average vehicle density $\overline{\mu_d}$ in all segments of the freeway stretch, that is, the standard deviation, so it is expressed as in Eq. (17) follows.

$$\begin{aligned} \sigma_d &= \sqrt{\frac{1}{J} \sum_{j=1}^{J} \left(\mu_{d_j} - \overline{\mu_d}\right)^2} \\ &= \sqrt{\frac{\sum_{j=1}^{J} \sigma_{d_j}^2}{J}} \end{aligned} \tag{17}$$

Therefore, combining Eqs. (16) and 17, we get the formula for waiting time STD σ_{T_J} as a function of average traffic flow $\overline{A}$, average vehicle density $\overline{\mu_d}$ and vehicle density heterogeneity σ_d shown in Eq. (18). It can be observed that through mathematical derivation, we have isolated the density heterogeneity term from the equation, which, as per our previous analysis, is also the fundamental reason for higher travel time fluctuations during congestion offset periods.

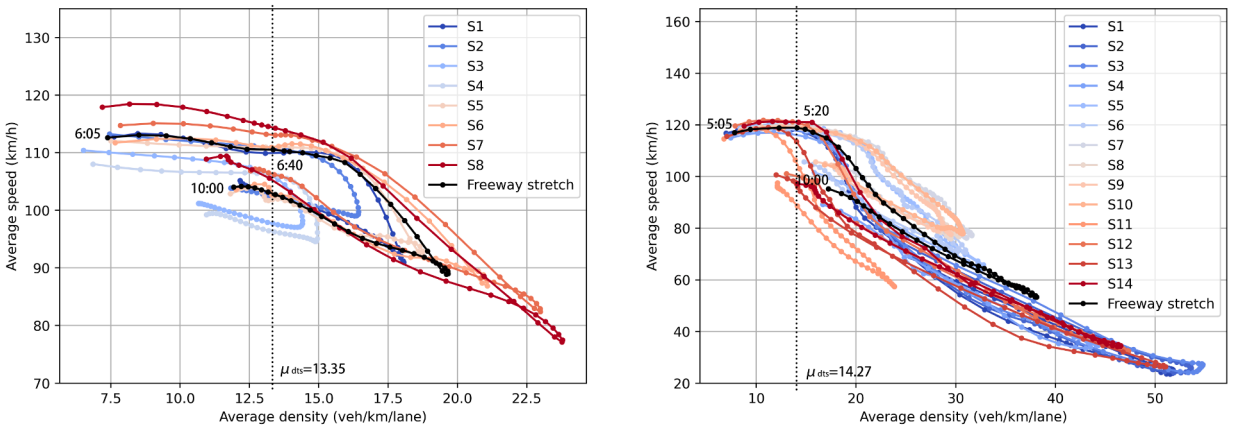
$$\sigma_{T_J} = \frac{l \cdot G}{\overline{A}} \sqrt{\zeta_1 \cdot \left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2 + \zeta_2 \cdot J \cdot \sigma_d^2 + \zeta_0} \tag{18}$$

Consequently, we square both sides of Eq. (18) before we proceed to fit this model using real-world data. This alteration does not impact the model's robustness, but facilitates the estimation of relationships between the variance of experienced travel time, average vehicle density, and density heterogeneity. It's worth noting that, in reality, when the freeway stretch is in a state of free flow and the average density $\overline{\mu_d}$ is below a certain density threshold μ_{dts}, it doesn't significantly affect the experienced travel time and its reliability. Therefore, the model needs to account for the disparity between the actual average density $\overline{\mu_d}$ and the density threshold μ_{dts}.

To establish the density threshold μ_{dts}, we analyzed the correlation between the average speed and average density of the selected freeway stretches I-694 and I-805, as demonstrated in Fig. 9.

As per Fig. 9, for the entire freeway stretch (black curves) and all its constituent segments, the average traffic speed remains virtually constant when the average density $\overline{\mu_d}$ is less than a specific μ_{dts}, which is determined by its capacity. Within this range, the vehicle speed isn't influenced by an increase in vehicle density and remains at free flow, with the TTV equaling zero. We observe that the density threshold μ_{dts} is around 13.35 veh/km/lane occurred at 6:40 am for freeway stretch I-694. For freeway stretch I-805, we add data from 5:00 am to 6:00 am because the experienced travel time STD increased notably from the onset of the morning peak period as shown in Fig. 2(a), implying that traffic speed is already influenced by the average vehicle density before 6:00 am. The density threshold μ_{dts} occurs at an earlier time, 5:20 am, with a value of around 14.27 veh/km/lane. The model is represented in Eq. (19).

$$\sigma_\tau^2(t) = \left(\frac{l \cdot G}{\overline{A}(t)}\right)^2 \cdot \left(\zeta_1 \cdot (J + 2\bar{r} \cdot (J-1)) \cdot (\overline{\mu_d}(t) - \mu_{dts})^2 + \zeta_2 \cdot J \cdot \sigma_d^2(t) + \zeta_0\right) \tag{19}$$



(a) I-694 (b) I-805

Fig. 9. Relationship between the average speed and average density for the selected freeway stretches I-694 and I-805 (all segments show a hysteresis process).

Here, $\sigma_\tau^2(t)$ denotes the experienced TTV at departure time interval t, $\overline{A}(t)$ represents average traffic flow, and $\overline{\mu_d}(t)$ stands for average density. $\sigma_d(t)$ represents density heterogeneity, $\bar{r}$ denotes the average Pearson correlation parameters between all adjacent segments, J is the number of segments in the freeway stretch, G signifies the number of lanes in the freeway stretch, and l is the length of each segment.

Furthermore, for the density heterogeneity term $\sigma_d(t)$ in Eq. (19), we model it based on the *birth and death process* of the M/M/1 model, that is, using the average density and flow of the freeway stretch to simulate the temporal change in density heterogeneity. While reducing the number of variables, this does not affect the form of Eq. (19), and the specific details are shown in Appendix I. The models Eqs. (19) and (I.7) correspond to scenarios without and with density heterogeneity modeling, respectively.

4. Results

In the context of queueing theory-based M/M/1 model, after enhancing the model by incorporating spatial density heterogeneity across different segments, we estimate the parameters for the two proposed models Eqs. (19) and (I.7). Furthermore, the models will be validated across various temporal and spatial conditions.

4.1. Model estimation

Based on real-world datasets, we estimate the parameters of the proposed models Eqs. (19) and (I.7) for freeway stretches I-694 and I-805. These two models have similar mathematical forms and parameters, except that Eq. (I.7) additionally considers modeling the density heterogeneity term $\sigma_d(t)$. Here, we employ the nonlinear least squares (NLS) method (Srinivasan and Mason, 1986) for parameter estimation. This method seeks the optimal values of model parameters by minimizing the sum of squares of the differences between the predicted values of the model and the actual observations, implemented using the SciPy library in Python. For freeway stretches I-694 and I-805, there are 41 and 48 data points respectively for estimation considering that the density of some data points in freeway stretch I-694 are less than the density threshold μ_{dts}. To illustrate the importance of density heterogeneity in explaining hysteresis loops, we add a model based on formula Eq. (5) that ignores density heterogeneity (σ_d^2) to the comparison with the other two proposed models, which considers the impact of average density and flow on travel time fluctuations. Additionally, we select two traditional benchmark models (Torrisi et al., 2017; Zang et al., 2022), namely a polynomial model and an exponential model, to fit the relationship between average density and travel time variance, for comparison with all our proposed models based on the M/M/1 queueing framework. The results for freeway stretches I-694 and I-805 are shown in Table 3 and Fig. 10.

From Fig. 10, we observe that the two benchmark models (fuchsia and cyan lines) show similar performance. They can describe the near-linear relationship between average density and travel time variance but fail to replicate the hysteresis loops, as they only account for the effect of density on travel time fluctuations. In contrast, in our proposed models, we observe that both Eq. (19) (red line) and Eq. (I.7) (green line) can well fit the counterclockwise hysteresis loop phenomenon appeared in the real data, and are better than Eq. (5) (yellow line) that does not consider density heterogeneity σ_d^2. This is more obvious for freeway stretch I-694, because the impact of density threshold μ_{dts} is more significant. This means that the model after further calibration has good performance even if only the first order covariance is considered. The results presented in Table 3 reveal that the root mean square error (RMSE) and R^2 metrics of model goodness of fit of model Eq. (19) for freeway stretches I-694 and I-805 are 0.184 and 0.952, and 4.186 and 0.956 respectively. In addition, as the length of the freeway stretch increases, the significance of the error term ζ_0 diminishes, consistent with the discussion of errors caused by ignoring small terms in the derivation of Eq. (15). Based on RMSE and R^2, we observe that Eq. (19) performs better than Eq. (5), which can be regarded as a single-segment queueing baseline because it captures the basic

Table 3
Comparison of TTR models between without, observed, and modeled density heterogeneity.

Parameters			ζ_1	ζ_2	ζ_0	μ_{dts}	RMSE	R^2
Polynomial model	I-694	–	–	–	–	–	0.624	0.609
	I-805	–	–	–	–	–	12.213	0.611
Exponential model	I-694	–	–	–	–	–	0.608	0.613
	I-805	–	–	–	–	–	12.407	0.608
Without σ_d^2 (Eq. (5))	I-694	–	–	–	–	–	0.586	0.620
	I-805	–	–	–	–	–	7.865	0.826
Observed σ_d^2 (Eq. (19))	I-694	Coefficient	−0.0111	0.0609	−1.068	13.237		
		F-statistic	63.971	143.963	63.975	60.638	0.184	0.952
		P-value	6.67e-10[a]	5.44e-15*	6.66e-10[a]	1.30e-9[a]		
	I-805	Coefficient	2.26e-3	0.0251	2.892	15.142		
		F-statistic	86.382	111.390	24.554	24.561	4.186	0.956
		P-value	3.96e-12[a]	7.12e-14[a]	1.02e-5[a]	1.02e-5[a]		
Modeled σ_d^2 (Eq. (I.7))	I-694	Coefficient	−5.03e-3	55.532	−14.249	12.803		
		F-statistic	62.447	225.599	185.412	11.491	0.155	0.966
		P-value	9.04e-10[a]	1.11e-16[a]	1.11e-16[a]	1.55e-3[a]		
	I-805	Coefficient	−6.64e-3	91.849	−49.383	13.730		
		F-statistic	70.016	154.191	17.351	16.183	3.711	0.969
		P-value	8.54e-11[a]	2.22e-16[a]	1.35e-4[a]	2.12e-4[a]		

[a] indicates statistical significance at the 0.01 level.

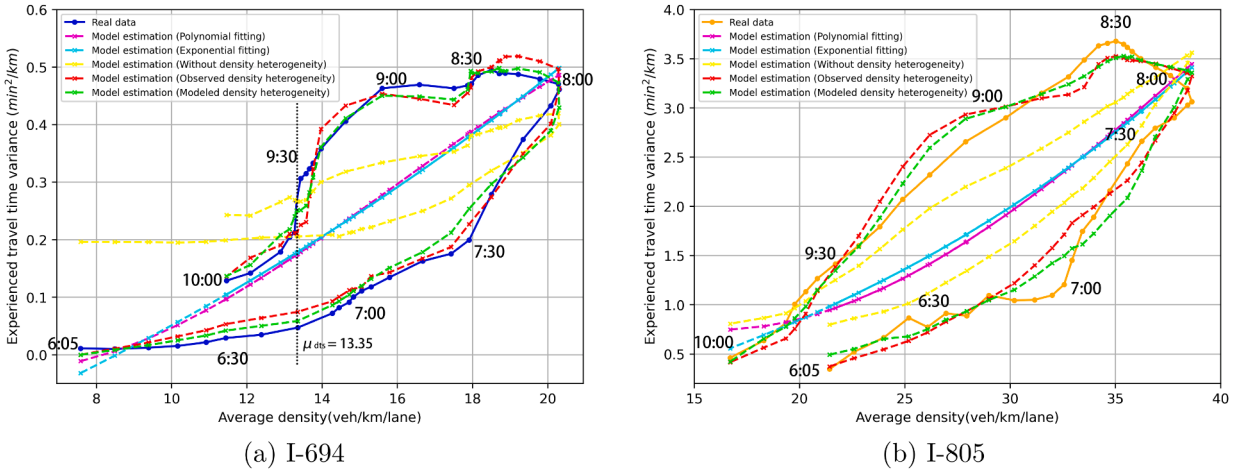


Fig. 10. Comparison between the estimation of the three proposed models, the two traditional benchmark models, and the real data of the experienced TTV.

flow-density-travel time variability relationship but ignores spatial density differences across segments. This improvement highlights the incremental benefit of the proposed multi-segment formulation: by incorporating density heterogeneity and adjacent-segment covariance, the model captures the spatial structure of congestion that single-segment or average-density-based models miss, and therefore more accurately describes the hysteresis loop phenomenon.

Moreover, we note that for freeway stretches I-694 and I-805, the density threshold μ_{dts} can be effectively estimated, and this value is around 14 veh/km/lane. For data smaller than this value, we use linear approximation. For freeway stretch I-694, the estimated value of μ_{dts} is equal to 13.24 veh/km/lane, which is consistent with the actual value of 13.35 veh/km/lane. In addition, the TTV σ_τ^2 has a positive correlation with density heterogeneity σ_d, but a negative correlation with average density $\overline{\mu_d}$. This goes against expectations because the average density $\overline{\mu_d}$ and density heterogeneity σ_d are not independent of each other, which is one reason why we need to perform additional modeling of density heterogeneity. Nonetheless, overall, the model estimates the impact of average density and density heterogeneity on day-to-day travel time reliability well when the freeway stretch is congested.

According to the RMSE and R^2 of the model shown in Table 3, Eq. (I.7) performs better than Eq. (19) for both freeway stretches, which provides a good estimate of the experienced TTR based on freeway stretch traffic flow and average vehicle density. According to the form of Eq. (I.7) and the results of the estimated parameters shown in Table 3, the TTV decreases with the increase of average traffic flow on the freeway under the same average density. Based on the relationship between flow, density, and speed, the faster the vehicle travels, the better the TTR (lower the TTV per km) will be. In addition, the TTV increases with the increase of the average vehicle density and vehicle density heterogeneity with the same traffic flow, which means that the slower the vehicle travels and more

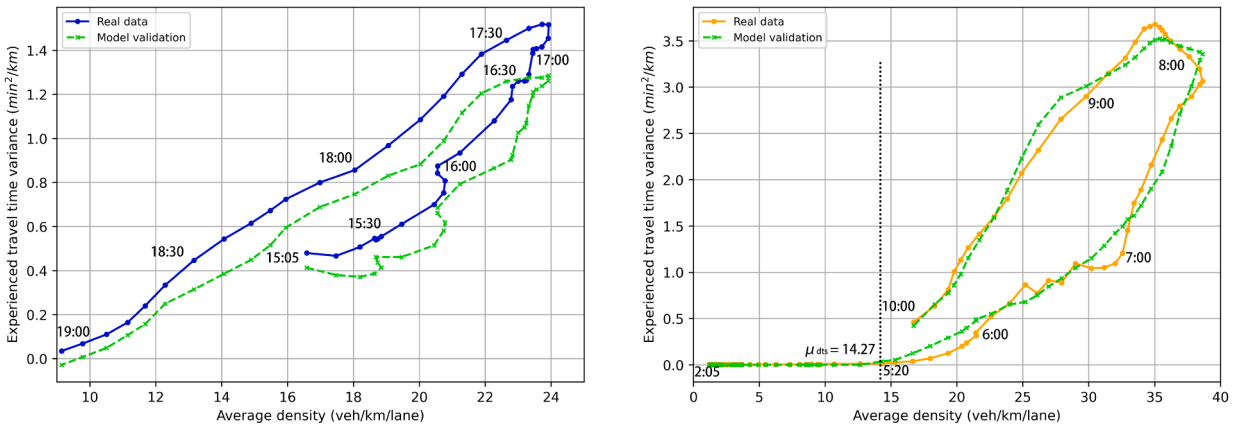


Fig. 11. Model validation for the two selected freeway stretches on other time ranges.

Table 4
Key information and validation results of the selected validation freeway stretches and subnetworks.

	Name	Length (km)	No. of detectors	No. of lanes	$\bar{r}$	RMSE	R^2
MSP freeway network	I-94 (N-S)	10.38	14	3	0.82	0.864	0.851
	I-394 (W-E)	5.82	8	3	0.78	0.407	0.824
	US-169 (N-S)	4.74	7	3	0.77	0.512	0.811
	Subnetwork	94.31	128	3	0.81	7.689	0.835
San Diego freeway network	I-5 (S-N)	15.87	14	4	0.88	6.581	0.842
	SR-163 (N-S)	13.97	9	4	0.84	5.637	0.831
	I-8 (E-W)	8.48	8	4	0.87	3.817	0.829
	Subnetwork	78.39	79	4	0.85	17.334	0.838

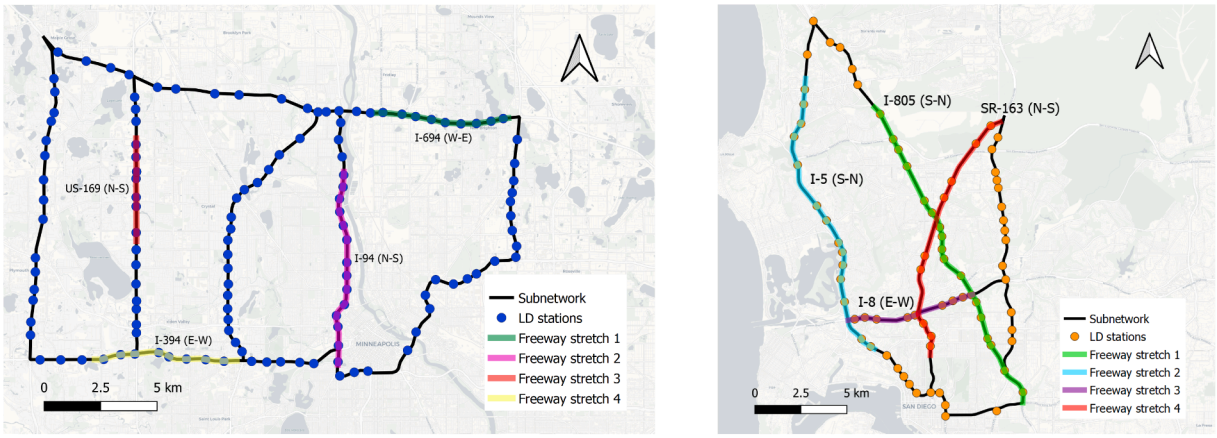
uneven the congestion distribution, the worse the TTR will be. From the relationship between the variables reflected by Eq. (I.7), in addition to the core problem studied in this paper, that is, the mechanism between density heterogeneity and TTR, the other two points can be confirmed in daily experience, which further illustrates the reliability of the model.

Overall, the proposed model better captures the nuanced temporal evolution of TTV with density, because it accounts for not only average density but also spatial density distribution and segment-level dependence. This provides a key explanation for its improved performance over simpler benchmark models and the single-segment queueing baseline. Next, we will validate the stability of the model over different temporal ranges and spatial locations.

4.2. Model validation across time and space

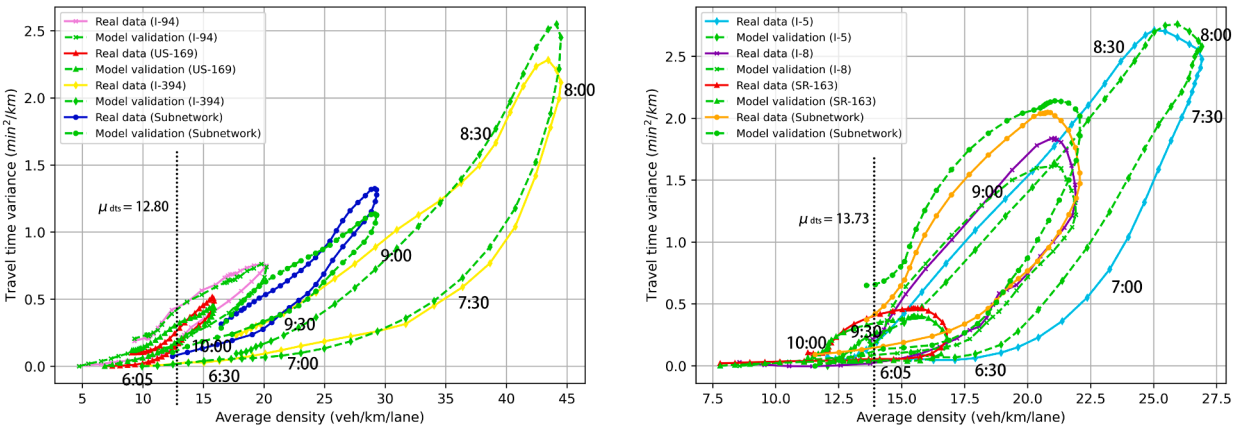
In this section, the model is validated using independent datasets in both temporal and spatial dimensions. Temporal validation uses different time periods from the same freeway stretches, while spatial validation uses other freeway stretches within the same subnetworks and the subnetworks containing all selected stretches. The parameters estimated from the morning peak data in Table 3 are kept fixed, and no validation observations are used for parameter estimation. First, in order to validate the robustness of the above model, we verify the model shown in Eq. (I.7) for two selected freeway stretches I-694 and I-805 in other time ranges except for the morning peak period. For freeway stretch I-694, we select the data of the time period from 3:00 pm to 7:00 pm during the evening peak period. For freeway stretch I-805, since there is no obvious evening peak, whose daily average traffic flow is lower than 1300 veh/hr/lane after 10:00 am, and no obvious density threshold μ_{dts} appears in the hysteresis curve of the morning peak period of 6:00 am-10:00 am, we select the data of the midnight period from 2:00 am to 6:00 am to validate the model and the estimation of the parameter μ_{dts}. The results are shown in Fig. 11.

As per Fig. 11, for freeway stretch I-694, our model successfully reproduces the counterclockwise hysteresis curve observed during the evening peak and accurately reflects the changes in experienced TTV. This is further substantiated by the RMSE and R^2 values of 0.274 and 0.853, respectively, implying the model's parameters are temporally robust. The discrepancy between the validation results and real data is mainly seen in the error term ζ_0. This is because during the evening peak, traffic flow and density are generally lower than during the morning peak. Consequently, traffic flow smoothness and density heterogeneity differ between these two periods. The evening peak sees variations between individual segment flow or density and the average flow or density of the freeway stretch ranging between 15% - 30% and 10% - 20%, respectively. These values contrast with those observed during the morning peak, leading to different approximation errors and omissions reflected in the error term ζ_0. This discrepancy is especially pronounced during busier times.



(a) MSP subnetwork (b) San Diego subnetwork

Fig. 12. Two extracted subnetworks for model validation.



(a) MSP freeway network (b) San Diego freeway network

Fig. 13. Comparison between model validation results and real data.

For freeway stretch I-805, the model accurately estimates the changes in experienced TTV during the early morning period (RMSE = 1.837 and $R^2 = 0.915$). This is based on data from 5:20 am to 6:00 am, when the average vehicle density exceeds the density threshold μ_{dts}. The model's estimated density threshold, μ_{dts}, aligns closely with the actual value recorded at 5:20 am (14.27 veh/km/lane), with an estimated result of 13.73 veh/km/lane as indicated in Table 3. This suggests that if the average vehicle density is below μ_{dts}, the experienced TTV does not drastically increase with an upsurge in vehicle density and density heterogeneity. Hence, the validation results demonstrate the model's effectiveness in representing changes in the TTR of freeway stretches over different time spans.

Next, we validate the model's generality in space. Given that morning peak hours exhibit more pronounced hysteresis loops compared to evening peak hours, we select the morning peak period to verify the model in other spatial locations. We extract two subnetworks from the MSP freeway network and the San Diego freeway network, which include the studied freeway stretches. These subnetworks contain 128 and 79 evenly distributed loop detector (LD) stations, respectively, along with three additional validation freeway stretches of varying occupancies and lengths, as shown in Fig. 12.

From Fig. 12, we observe that each of the two subnetworks contains four freeway stretches, with the first one being the freeway stretch studied in this paper, and the remaining three are validation freeway stretches. In addition, we extend the validation to the entire subnetwork to assess the model's generalization ability at the network level. Note that at the network level, the travel time is considered as instantaneous travel time rather than experienced travel time (Yildirimoglu et al., 2015). Therefore, we collected data from all detectors within the subnetworks from 6:00 am to 10:00 am during the morning peak in 2018. For model Eq. (I.7), the parameters such as number of segments J, number of lanes G, length of each segment l, and average Pearson correlation parameters $\bar{r}$ need to be updated since the validation freeway stretches and subnetworks have different spatial characteristics. Based on the

estimation results of model parameters in Table 3, the key information of the validation freeway stretches and subnetworks and the corresponding validation results are shown in Table 4, and the comparison between the model and the real data is shown in Fig. 13.

From Table 4 and Fig. 13, we observe that the model performs well across freeway stretches of varying occupancies and lengths, and also demonstrates strong generalization in predicting travel time fluctuations at the network level. Additionally, we note that in the TTV results of the MSP and San Diego freeway networks, the same average density corresponds to significantly different TTVs. There are two reasons for this. First, the average traffic flow during peak periods in the San Diego network is about 10% lower than that in the MSP freeway network, meaning that under the same density conditions, the speed is lower and congestion is more severe. Second, according to Table 4, we observe that the first order average Pearson correlation parameter $\bar{r}$ in the San Diego network is slightly higher than in the MSP freeway network. Based on previous discussions, this leads to more significant travel time fluctuations. Therefore, combined with Eq. (I.7), we know that these two factors result in higher TTV. In addition, the density threshold μ_{dts} remains consistent across different regions of the same network. Overall, the model validation results show that the model has good spatial generalization performance because different locations in the freeway network exhibit similar traffic flow and congestion patterns during the same peak hour (Gao and Levinson, 2024a; Zhang et al., 2016). Therefore, our proposed model demonstrates robustness and generalization in both time and space.

5. Conclusions

This paper advances the research conducted in Gao and Levinson (2021) regarding the time-lag relationship between occupancy heterogeneity and day-to-day travel time reliability (TTR) on the Minneapolis-St. Paul freeway network. It introduces, for the first time, an exploration into the effects of density heterogeneity on TTR, while also presenting a robust and comprehensive model for calculating the variance of experienced travel time on freeway stretches based on the M/M/1 queueing model. This model successfully simulates and explains the hysteresis phenomenon.

In Section 2.1, we selected freeway stretches I-694 (W-E) from the Minneapolis-St. Paul network and I-805 from the San Diego network for our data analysis. We leveraged the half-segment (HS) method to obtain accurate travel times, thus overcoming the significant underestimation inherent in the loop detector method. The resulting relationship curves between average vehicle density, experienced travel time standard deviation, and vehicle density heterogeneity verified the presence of a counter-clockwise hysteresis loop phenomenon. Moreover, it was determined that the effect of vehicle density heterogeneity on TTR becomes more significant as congestion moderates, as discussed in Section 2.2.

In Section 3.1, we identified that a time-homogeneous Poisson process with varying arrival rates A can effectively depict the vehicle arrival process for segments with or without on-ramps at different departure time intervals. This was validated by the exponential distribution of vehicle arrival intervals at different times during the morning peak period. Furthermore, it was established that the travel time for segments, irrespective of the presence or absence of on-ramps, can be fitted by the exponential distribution with parameter $D - A$. Therefore, both types of segments satisfy the basic characteristics of the M/M/1 queueing model.

Continuing in Sections 3.2–3.4, we validated the existing theories about the nearly linear relationship between travel time variability and mean travel time based on the simple M/M/1 queueing model, and derived that as travel distance increases, the parameter describing this linear relationship will decrease. Furthermore, we also determined, through an analysis of the travel time correlations among different segments of the freeway stretch, that these correlations are higher during congestion offset periods. This leads to higher covariance among the travel times of different segment pairs and higher total travel time STD under the condition of constant total travel time. The reason for this is the increased density heterogeneity during offset periods. Then, we developed a TTR model for freeway stretches consisting of multiple segments. Initially, based on the fundamental properties of the M/M/1 queueing model, we derived a model for TTV, average vehicle density, and vehicle density heterogeneity. This model factors in the correlation of travel time between segments and the density threshold μ_{dts} of the freeway stretch. During the derivation process, it was noted that discrepancies in the arrival traffic flow A across segments introduce errors into the model. Subsequently, we introduced another model accounting for the interdependence of two non-independent variables, the average vehicle density and density heterogeneity. Ultimately, by integrating these two models, we proposed a model linking experienced TTV with average vehicle density.

In Sections 4.1 and 4.2, we estimated the parameters of the models, and validated the robustness and generality of our model both temporally and spatially with previously estimated parameters. Temporally, we used previously obtained parameters to verify the TTR during the evening peak period for freeway stretch I-694, yielding an R^2 value of 0.853. For freeway stretch I-805, we validated the TTR during the early morning period, achieving an R^2 value of 0.915. Spatially, we processed the morning peak data from the two selected subnetworks and the three additional validation freeway stretches they contain to validate the model's spatial generalization ability. These results confirmed the model's robustness and generality in both time and space.

For the real-world application of the model proposed in this paper, we consider two aspects: real-time prediction of freeway network travel time variability (TTV) and control strategies to improve network travel time reliability (TTR). Compared with existing TTR methods that mainly rely on average traffic states or empirical curve fitting, the proposed model provides a more mechanism-based and spatially explicit representation of TTV. Its practical advantage lies in distinguishing traffic states that have similar aggregate conditions but different spatial congestion patterns, thereby offering more informative inputs for both reliability prediction and control-oriented decision-making. For real-time prediction of TTV, our model indicates that, in addition to the adverse effects of congestion caused by the rise in average density from a traditional perspective, higher spatial density heterogeneity also negatively impacts TTV by increasing the correlation between the travel times of different segments. Therefore, whether improving traditional mathematical models (Torrisi et al., 2017; Van Lint and van Zuylen, 2005) or developing emerging deep learning models, such as neural networks (Shao et al., 2024; Yang et al., 2009), it is essential to include density heterogeneity and its effects as input features.

For traffic control measures aimed at improving freeway network TTR, the findings in this paper can also provide valuable guidance. This is based on the recognition of the negative impact of density heterogeneity on TTR under equivalent average densities. For instance, while implementing common ramp and perimeter controls (Ramezani et al., 2015; Hegyi et al., 2005), we posit that establishing density homogeneity as an optimization goal can help achieve lower density heterogeneity at the same average density. As a result, TTR performance is improved without losing capacity, also enhancing the overall network capacity (Geroliminis and Sun, 2011a; Gao and Levinson, 2024b). For example, two freeway corridors may have the same average density but different spatial density distributions. Traditional average-density-based TTR models would tend to predict similar reliability levels, whereas the proposed model predicts higher TTV for the corridor with stronger density heterogeneity. This distinction can support more targeted control decisions, such as applying coordinated ramp metering or variable speed limits to over-congested segments to smooth the density distribution and improve network-level TTR.

The following issues should be considered in future research.

- First, compared to travel time data based on loop detectors, real travel trajectory data can better reflect the travel time fluctuations due to the uneven distribution of vehicle density.
- Secondly, the model is derived based on the M/M/1 queueing model of a single server, that is, without considering lane changing, a perhaps more accurate M/M/n queueing model with multiple servers that could be developed recognising that lane changing has a significant impact on travel time and reliability.
- Third, the applicability of the proposed model to urban street networks remains to be examined. Unlike freeways, urban networks are affected by signals, intersections, turning movements, route choices, multimodal interactions, and spillbacks. These features may require extensions of the current M/M/1-based framework, such as signal-dependent service rates, intersection-level queues, and topology-dependent congestion propagation.
- Fourth, density heterogeneity may also be affected by weather, incidents, and heterogeneous driving behavior, as these factors can reduce local capacity, disturb arrival and departure processes, and create uneven congestion accumulation across segments. Although such effects are partly absorbed by the error term ζ_0, future studies could use richer datasets to better isolate their impacts on TTR.
- Fifth, the persistence of the hysteresis loop under non-recurrent conditions should be further tested. Special events, adverse weather, incidents, or work zones may alter demand, reduce local capacity, and intensify spillback, thereby enlarging or reshaping the hysteresis loop. Event-day, rainy-day, or incident-period datasets could be used to examine its robustness and implications for network-level TTR.

CRediT authorship contribution statement

Yang Gao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **David Levinson:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Half-segment (HS) method

For each segment on the freeway stretch, we assume that each detector covers only half the distance of its adjacent segments, which means that the loop detector located on the freeway stretch edge is considered to cover half of its corresponding segment, and the remaining loop detectors are considered to cover half of its upstream and downstream segments. The distance here is the exact value determined from their GPS coordinates data. The half-segment (HS) method proposed in this paper and the traditional constant speed interpolation (CSI) method are shown in Eqs. (A.1) and (A.2) respectively.

$$\tau_{HS}(t) = 0.5\left(\frac{l}{V_{up,t}} + \frac{l}{V_{down,t}}\right) \tag{A.1}$$

$$\tau_{CSI}(t) = \frac{2l}{V_{up,t} + V_{down,t}} \tag{A.2}$$

Where $\tau_{CSI}(t)$ and $\tau_{HS}(t)$ are the travel time of the segment calculated by constant speed interpolation method and half-segment method respectively, $V_{up,t}$ and $V_{down,t}$ are the speed detected by its upstream and downstream loop detectors at departure time interval t respectively, and l is the distance between these two consecutive loop detectors. The difference of the segment travel time measured by these two methods are shown in Eq. (A.3).

$$\Delta_{\tau_{HS}(t)-\tau_{CSI}(t)} = \frac{0.5l(V_{up,t} - V_{down,t})^2}{(V_{up,t} \cdot V_{down,t}) \cdot (V_{up,t} + V_{down,t})} \tag{A.3}$$

Through Eq. (A.3), we see that $\tau_{HS}(t)$ is always greater than or equal to $\tau_{CSI}(t)$, and when the speed gap between the two detectors becomes larger, the difference between $\tau_{HS}(t)$ and $\tau_{CSI}(t)$ also becomes larger, because the value of the numerator becomes larger and the value of the denominator becomes smaller when the sum of the speed of these two loop detectors is unchanged. This is more in line with the actual situation because more detector speed gaps result in more severe travel time underestimation caused by acceleration and deceleration. Therefore, the travel time obtained by the HS method is more accurate, which better offsets the underestimation of the travel time and the $\mu_\tau(t)$ appearing in this paper are all obtained by the half-segment (HS) method.

Appendix B. Formulas involved in hysteresis loop phenomenon

The formulas for the experienced travel time mean and STD are shown in Eqs. (B.2) and (B.3) respectively.

$$\mu_{\tau,mJ}(t) = \sum_{j=0}^{J-1} \tau_{m(j+1)}(t + \mu_{\tau,mj}(t)) \tag{B.1}$$

$$\mu_\tau(t) = \frac{1}{M} \sum_{m=1}^{M} \mu_{\tau,mJ}(t) \tag{B.2}$$

$$\sigma_\tau(t) = \sqrt{\frac{1}{M-1} \sum_{m=1}^{M} (\mu_{\tau,mJ}(t) - \mu_\tau(t))^2} \tag{B.3}$$

where t is one-minute or five-minutes departure time interval, M is the number of days, J is the number of segments in the freeway stretch and $\tau_{mj}(t)$ is the travel time within the segment j at departure time interval t on day m, $\mu_{\tau,mj}(t)$ is the experienced travel time of completing the route from the origin to the end of the segment j of the freeway stretch.

The formulas for the vehicle density mean and heterogeneity are shown in Eqs. (B.5) and (B.6) respectively.

$$\mu_{d,m}(t) = \frac{1}{J} \sum_{j=1}^{J} d_{mj,t} \tag{B.4}$$

$$\mu_d(t) = \frac{1}{M} \sum_{m=1}^{M} \mu_{d,m}(t) \tag{B.5}$$

$$\sigma_d(t) = \frac{1}{M} \sum_{m=1}^{M} \sqrt{\frac{1}{J} \sum_{j=1}^{J} (d_{mj,t} - \mu_{d,m}(t))^2} \tag{B.6}$$

where $\mu_{d,m}(t)$ is the average vehicle density of the freeway stretch at departure time interval t on day m, $d_{mj,t}$ is the vehicle density of segment j within the freeway stretch at departure time interval t on day m, which is equal to the average results of the two loop detectors corresponding to the upstream and downstream. It should be noted that the average vehicle density here is measured at an instantaneous departure time interval, and since the travel time of the selected freeway stretches are about 4 min and 12 min respectively, and the vehicle density has a strong continuity in time, therefore, for this small travel time, the instantaneous vehicle density can indicate the vehicle density during the entire driving process well, and correspond to experienced travel time obtained before.

Appendix C. Detailed discussion about Proposition 1

Remark 1. For the I-694 stretch, four segments each contain an on-ramp, and the Pearson correlation coefficients of their mainline flow and on-ramp flow are 0.871, 0.821, 0.825, and 0.862, respectively. As a result, it's unnecessary to consider the mainline flow and on-ramp flow as two independent variables when determining their sum's probability distribution. Instead, we regard their sum (the total traffic flow) as a single variable to validate its distribution. The distribution of day-to-day arrival time intervals for different time points for segments with and without on-ramps is obtained from their daily traffic flow, with arrival time intervals taking the average value of the corresponding segments. The results for I-694 are displayed in Fig. C.1, and the fitting results are summarized in Table C.1.

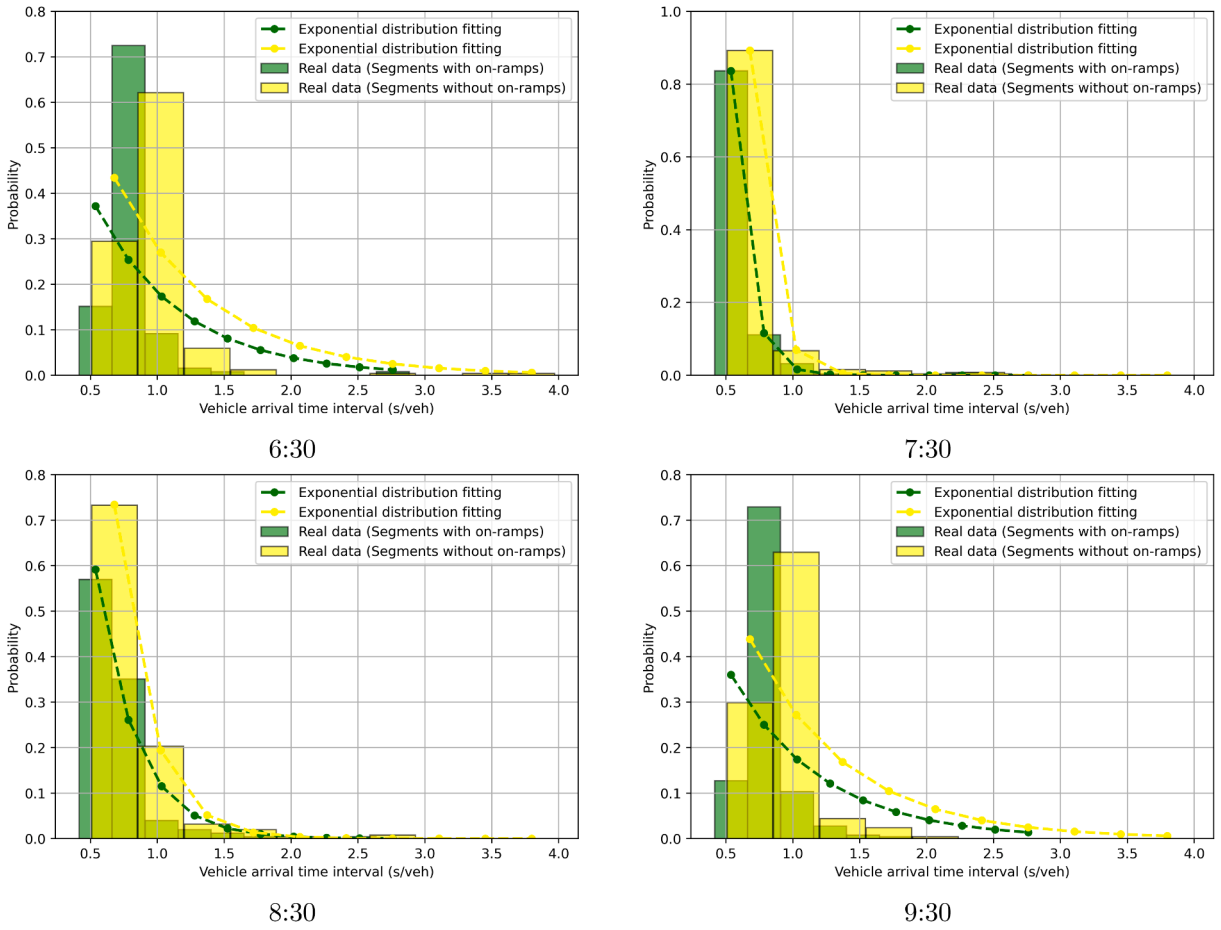


Fig. C.1. Vehicle arrival time interval distribution and exponential distribution fitting results for segments with on-ramps and segments without on-ramps of freeway stretch I-694 at four selected time points.

Table C.1
Fitting results of arrival time interval distribution for segments with on-ramps and segments without on-ramps of freeway stretch I-694.

	Segments with on-ramps				Segments without on-ramps			
Time points	6:30	7:30	8:30	9:30	6:30	7:30	8:30	9:30
Estimated A	1.55	8.00	4.48	1.47	1.38	7.36	3.84	1.38
R^2	0.55	0.99	0.96	0.53	0.68	0.99	0.98	0.71
Traffic flow (veh/h)	4208	5256	4667	4168	3384	4265	3787	3374
Pearson correlation coefficients	0.99				0.99			

As indicated in Fig. C.1, segments with on-ramps exhibit a slightly higher estimated arrival rate A compared to segments without on-ramps, attributable to on-ramp traffic flow. Moreover, it's noteworthy that the vehicle arrival time interval during different time points within the morning peak period can be effectively approximated by an exponential distribution for both segments with and without on-ramps, particularly during the busiest times of the morning peak period. The improved fitting and more pronounced hysteresis loop phenomenon during these peak times suggest a correlation between the arrival time interval adhering to the exponential distribution and the occurrence of the hysteresis loop phenomenon. Detailed exploration of how the distribution pattern influences this phenomenon can be addressed in future research. Nonetheless, the observation implies that both conditions meet the Poisson flow characteristics.

Appendix D. Detailed discussion about the relationship between travel time and vehicle arrivals

Given the challenges associated with collecting actual queueing times and service times for vehicles traversing the freeway stretch during peak hours, we can collect Pearson correlation coefficients between vehicle arrival intervals and service times from midnight to 10:00 am as shown in Fig. D.1. This approach allows us to assess whether these two distributions are independent during off-peak hours when travel times are approximately equal to service times, that is, travel times without congestion.

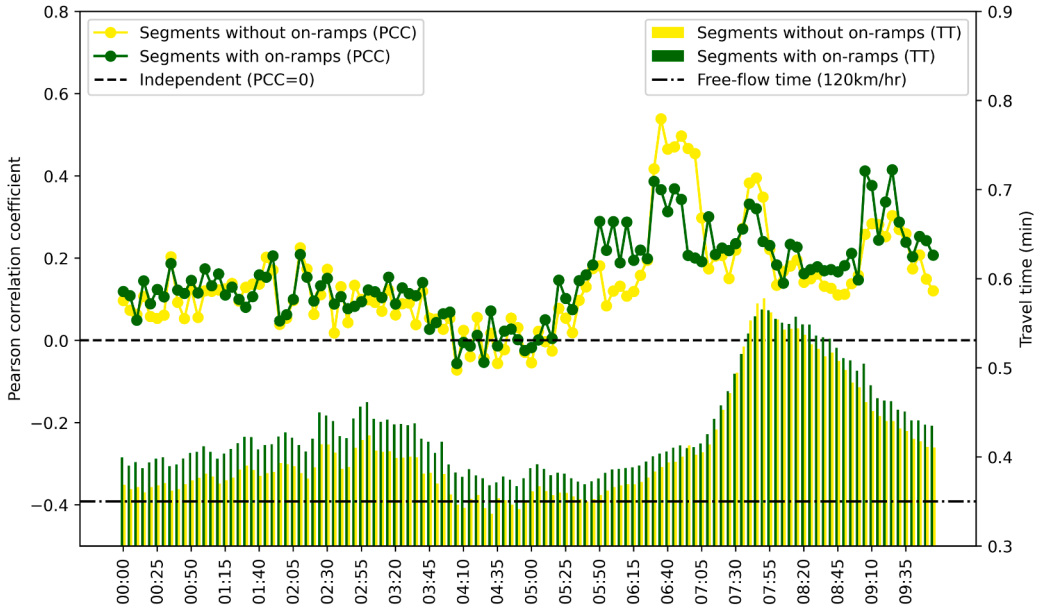


Fig. D.1. Pearson correlation coefficient variations between arrival time intervals and travel times from midnight to 10:00 am of freeway stretch I-694.

We can observe that around 4 am to 5:30 am, the vehicle travel times are approximately equal to the time spent in freeflow conditions, with travel speeds around 120 km/hr. During this period, the travel times, which represent the service times, are independent of the arrival intervals, satisfying the basic assumption of the M/M/1 model. In reality, the service time is influenced by factors such as road speed limits, characteristics of the vehicle, and other external conditions that are not related to congestion but are related to the freeway or the vehicle itself. Therefore, it is not directly correlated with the rate of vehicle arrivals. However, during other busier time periods, travel times tend to positively correlate with arrival intervals as congestion increases. This implies that queueing times, representing congested time, are correlated with arrival intervals, also aligning with the model's fundamental assumptions.

Appendix E. Detailed discussion about Proposition 2

Remark 2. Following the previous method, we obtain the day-to-day travel time distribution for segments with and without on-ramps at various time points during the morning peak period. The results are illustrated in Fig. E.1, and the fitting results are summarized in Table E.1.

Table E.1
Fitting results of travel time distribution for segments with on-ramps and segments without on-ramps of freeway stretch I-694.

	Segments with on-ramps				Segments without on-ramps			
Time points	6:30	7:30	8:30	9:30	6:30	7:30	8:30	9:30
Estimated $D-A$	14.13	6.93	9.48	14.28	14.31	7.17	10.92	14.53
R^2	0.99	0.97	0.94	0.99	0.99	0.96	0.98	0.99
Estimated D	15.68	14.93	13.96	15.75	15.69	14.53	14.76	15.91

As indicated in Fig. E.1, the exponential distribution fits the travel time during the morning peak period for both segments with and without on-ramps. When coupled with the estimated arrival rate A detailed in Table C.1, we can deduce the departure rate D, as summarized in Table E.1. It's observed that the departure rate D for segments with and without on-ramps does not exhibit a consistent relative magnitude, as travel time for segments without on-ramps can also be influenced by the bottlenecks induced by their adjacent

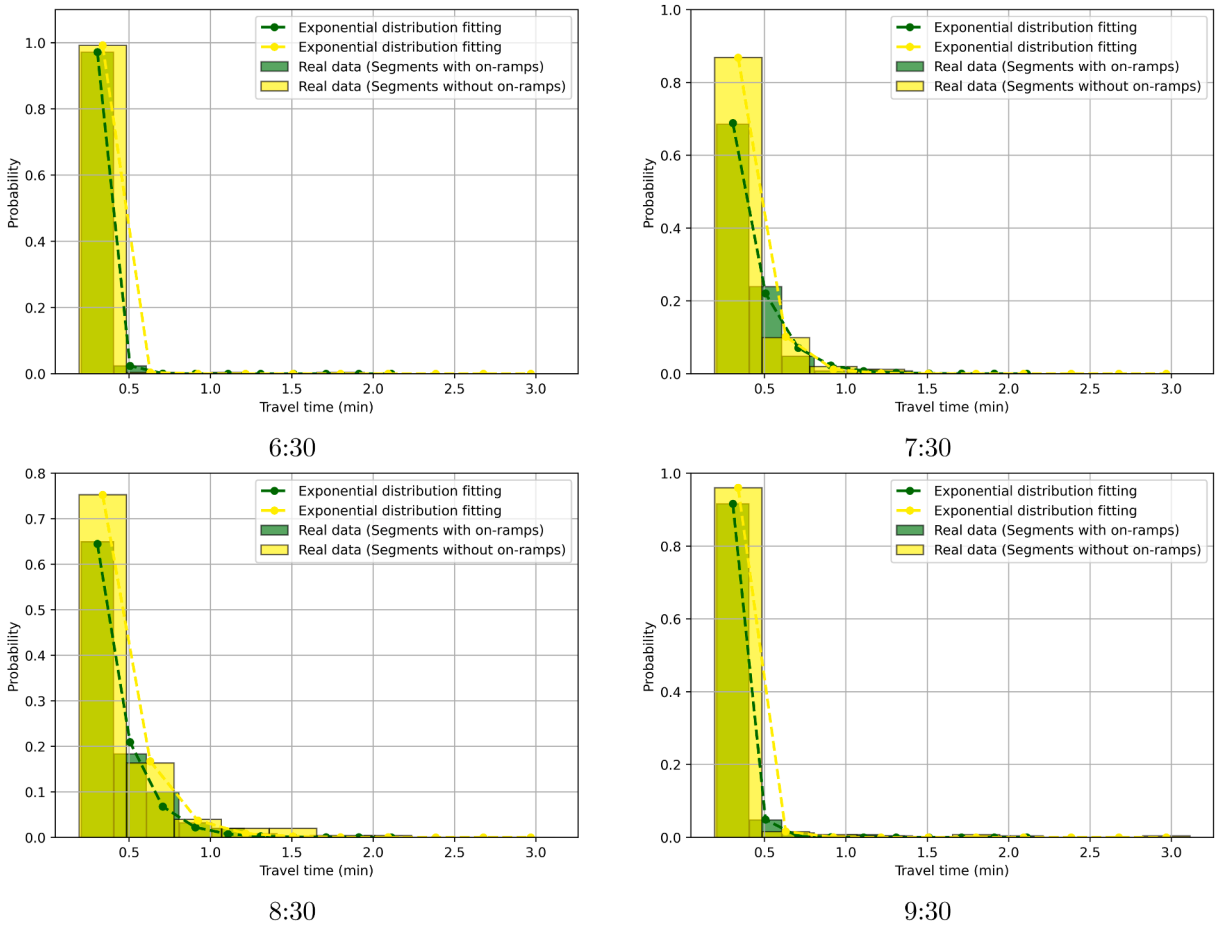


Fig. E.1. Vehicle travel time distribution and exponential distribution fitting results for segments with on-ramps and segments without on-ramps of freeway stretch I-694 at four selected time points.

segments with on-ramps. Compared to the arrival rate A, the fluctuation of the departure rate D is not substantial in time. As such, we can treat the departure rate D as a constant in the model. Furthermore, although the estimated result of the departure rate D being greater than the arrival rate A is inevitable, it is reasonable because the queue length on the freeway stretch will not grow indefinitely, meaning the model remains stable.

Appendix F. Correlation matrices

We have derived the Pearson correlation coefficient matrices for the travel time at four selected time points, as displayed in Figs. F.1 and F.2. Here, S1 through S8 correspond to the 8 segments of I-694, and S1 through S14 correspond to the 14 segments of I-805, arranged from upstream to downstream.

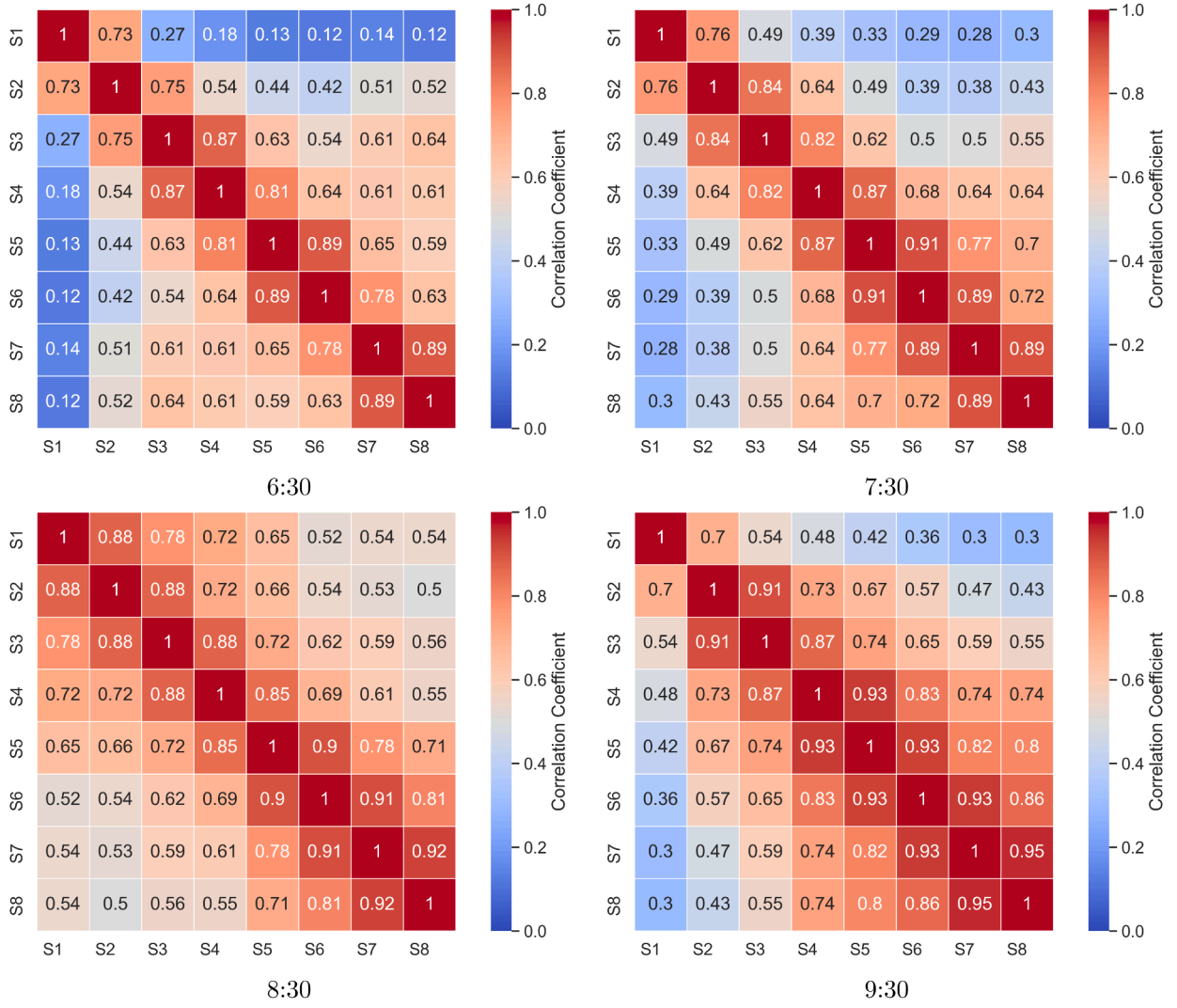


Fig. F.1. Pearson correlation coefficient matrix of travel time on freeway stretch I-694 at four selected time points.

As per Figs. F.1 and F.2, the travel time for a chosen segment shows a strong correlation with the travel times of other segments both at the onset and offset of the congestion, particularly those adjacent to it.

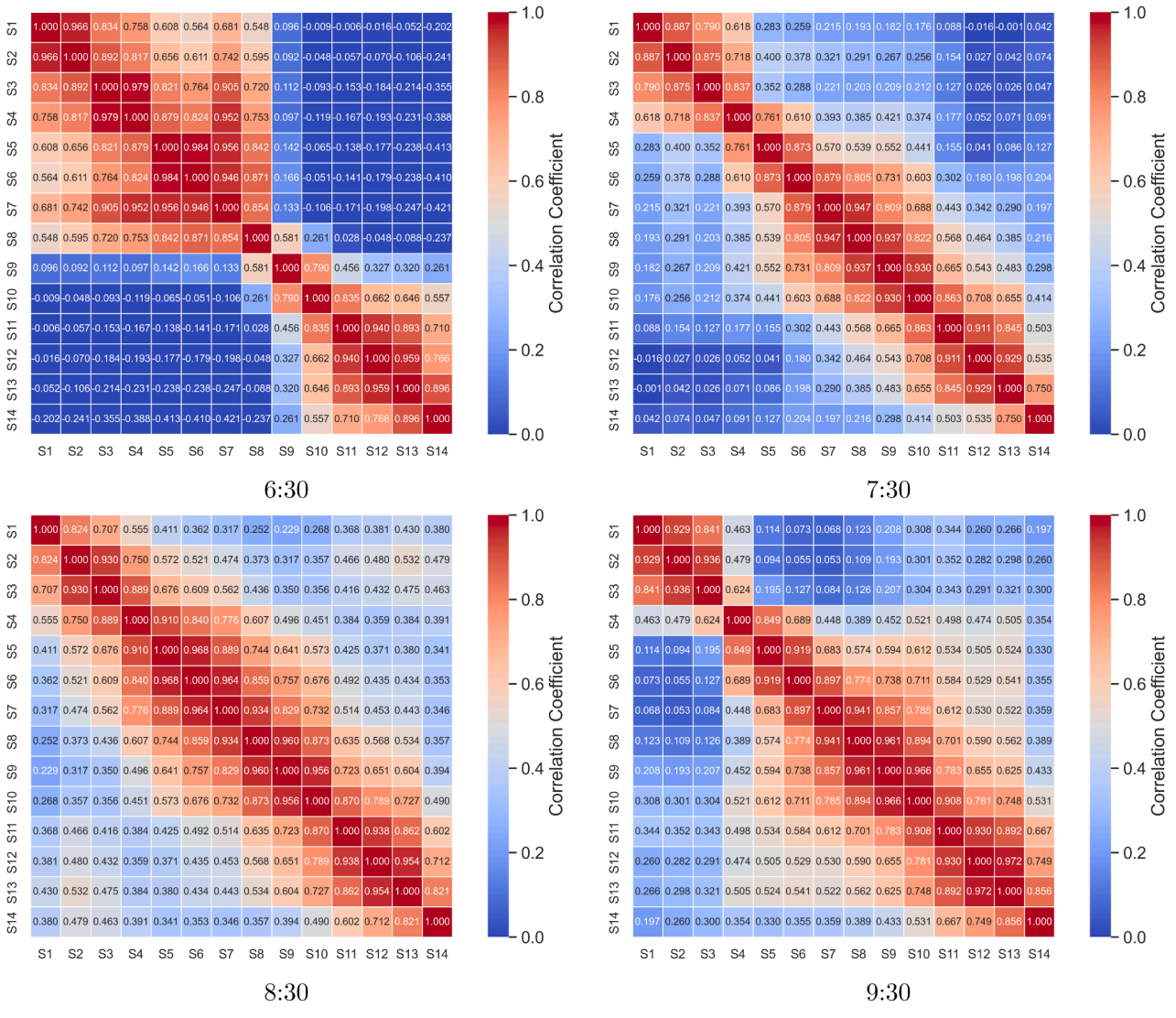


Fig. F.2. Pearson correlation coefficient matrix of travel time on freeway stretch I-805 at four selected time points.

Appendix G. Detailed proof about Proposition 4

Consider the expected travel time of J segments $E(T_1), E(T_2), \ldots, E(T_J)$ such that their sum, i.e. total travel time of the freeway stretch, is a constant, which means:

$$E(T_1) + E(T_2) + \ldots + E(T_J) = c \tag{G.1}$$

where c is a constant total travel time, which means under the same mean total travel time conditions.

We want to prove that the sum of the squares of the travel time of all segments, i.e. the first term $\sum_{j=1}^{J}(E(T_j))^2$ in Eq. (7), is minimized when all travel times are equal. To prove this, we can use the method of Lagrange multipliers. Let L be the Lagrangian function:

$$L(E(T_1), E(T_2), \ldots, E(T_J), \lambda) = E(T_1)^2 + E(T_2)^2 + \ldots + E(T_J)^2 + \lambda(c - E(T_1) - E(T_2) - \ldots - E(T_J)) \tag{G.2}$$

where λ is the Lagrange multiplier.

Now, we take partial derivatives of L with respect to expected travel time of each segment and set them equal to zero:

$$\frac{\partial L}{\partial E(T_1)} = 2E(T_1) - \lambda = 0,$$

$$\frac{\partial L}{\partial E(T_2)} = 2E(T_2) - \lambda = 0,$$

$$\vdots$$

$$\frac{\partial L}{\partial E(T_J)} = 2E(T_J) - \lambda = 0.$$

Solving these equations, we get $E(T_1) = E(T_2) = \ldots = E(T_J) = \frac{\lambda}{2}$. Then, substituting this back into Eq. (G.1) to solve the λ:

$$\frac{\lambda}{2} + \frac{\lambda}{2} + \ldots + \frac{\lambda}{2} = c$$

$$\lambda = \frac{2c}{J}$$

Then, substituting this back into Eq. (G.2) for the sum of the squares of the travel time of segments:

$$E(T_1)^2 + E(T_2)^2 + \ldots + E(T_J)^2 = \frac{\lambda^2}{4} + \frac{\lambda^2}{4} + \ldots + \frac{\lambda^2}{4} = J\left(\frac{\lambda^2}{4}\right) = \frac{c^2}{J}.$$

Since λ is a constant, therefore, the sum of the squares of the travel time of all segments, i.e. $\sum_{j=1}^{J}(E(T_j))^2$, is minimized when all travel times of segments on the freeway stretch are equal, i.e., $E(T_1) = E(T_2) = \ldots = E(T_J) = \frac{\lambda}{2} = \frac{c}{J}$. And this minimum value is $\frac{c^2}{J}$.

Appendix H. Minor term in Eq. (15)

$Ratio^1$ and $Ratio^2$ represent the ratio of the first item and the second item of the last term to the other terms shown in Eq. (15).

$$\begin{aligned} Ratio^1 &= \left| \frac{2\bar{r} \cdot \overline{\mu_d} \cdot (\sigma_{d_1} + \sigma_{d_J})}{\left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2 + \sum_{j=1}^{J} \sigma_{d_j}^2} \right| \\ &\leq \left| \frac{2\overline{\mu_d} \cdot (\sigma_{d_1} + \sigma_{d_J})}{\left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2} \right| \\ &= \left| \frac{2}{\frac{\left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}}{\sigma_{d_1} + \sigma_{d_J}}} \right| \end{aligned} \tag{H.1}$$

$$\begin{aligned} Ratio^2 &= \left| \frac{\bar{r} \cdot \left(\sum_{\substack{1 \leq j_1, j_2 \leq J, \\ |j_1 - j_2| = 1}} \sigma_{d_{j_1}} \cdot \sigma_{d_{j_2}} \right)}{\left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2 + \sum_{j=1}^{J} \sigma_{d_j}^2} \right| \\ &\leq \frac{2\sum_{j=1}^{J} \sigma_{d_j}^2 - \sigma_{d_1}^2 - \sigma_{d_J}^2}{\left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2 + \sum_{j=1}^{J} \sigma_{d_j}^2} \\ &\leq \frac{2}{\frac{\left(J + 2\bar{r} \cdot (J-1)\right) \cdot \overline{\mu_d}^2}{\sum_{j=1}^{J} \sigma_{d_j}^2} + 1} \end{aligned} \tag{H.2}$$

Appendix I. Modeling spatial density heterogeneity based on queueing theory

In the M/M/1 queueing model, the number of vehicles in the system satisfies the *birth-death process* (Newell, 2013), which is a special case of continuous-time Markov process and widely used in traffic modeling for travel time estimation, congestion prediction, etc. (Daganzo, 1978; Chakroborty, 2013; Banishree, 2020) Based on the condition that the system remains in equilibrium and the utilization rate ρ, the probability mass function (PMF) of the number of vehicles included in the freeway stretch is shown in Eq. (I.1).

$$P(n) = \rho^n(1-\rho) \tag{I.1}$$

According to Eq. (I.1), the partial derivatives of the probability when the number of vehicles existing in the freeway stretch is equal to n with respect to the arrival rate A and departure rate D are shown in Eqs. (I.2) and (I.3) respectively.

$$\frac{\partial P(n)}{\partial A} = \frac{n \cdot A^{n-1}}{D^{n+1}} \cdot (D-A) > 0 \tag{I.2}$$

$$\frac{\partial P(n)}{\partial D} = \frac{n \cdot A^{n}}{D^{n+2}} \cdot (A-D) < 0 \tag{I.3}$$

According to Eqs. (I.2) and (I.3), we notice that Eq. (I.2) is greater than 0, Eq. (I.3) is less than 0, because utilization rate $\rho = \frac{A}{D} < 1$, and if n is large enough, it means that the vehicle density in freeway stretch is dense enough to cause congestion, then, the probability of congestion in freeway stretch increases as the arrival rate A increases, and as the departure rate D decreases. This is consistent with the actual situation, when the traffic flow increases or the capacity of the freeway stretch drops, the freeway stretch is more prone to congestion. Therefore, *birth-death process* can well illustrate the relationship between the vehicle density and arrival rate A and departure rate D.

Combining Eq. (I.1) with the expectation of the number of vehicles on the freeway stretch shown in Eq. (3), the variance of the number of vehicles in the freeway stretch σ_n^2 is shown in Eq. (I.4).

$$\begin{aligned}
\sigma_n^2 &= \sum_{n=0}^{+\infty} \rho^n(1-\rho) \cdot (n - E(n))^2 \\
&= \sum_{n=0}^{+\infty} \rho^n(1-\rho) \cdot (n - \frac{\rho}{1-\rho})^2 \\
&= \frac{\rho}{(1-\rho)^2} \\
&= \frac{E(n)^2}{\rho} = \frac{D \cdot E(n)^2}{A} = \frac{D \cdot (L \cdot G \cdot \overline{\mu_d})^2}{A}
\end{aligned} \tag{I.4}$$

From Eq. (I.4), we find that the variance and the expectation of the number of vehicles on the freeway stretch satisfies a quadratic relationship, which is related to the arrival rate A and the departure rate D, but we need to note that the relationship described by Eq. (I.4) is based on the state probability space, not based on the actual space. Therefore, we need to map this relationship to the actual space. We have:

Proposition 6. *The spatial distribution of vehicle density approximately satisfies a birth-death process, and the variance of its distribution can be indicated by Eq. (I.4).*

Remark 6 We assume that the freeway stretch is divided into enough segments, and the number of vehicles in each segment is only related to the number of vehicles in its adjacent upstream segment, that is, continuous-time Markov process, and within a short time interval, can be regarded as time-homogeneous, since the distribution of vehicle density over the space on freeway stretch is continuous and strongly correlated, especially during congestion period (Yildirimoglu and Geroliminis, 2013; Chiabaut and Faitout, 2021). Therefore, we assume that the arrival rate A of any segment is the average traffic flow at time interval t on the freeway stretch, and the departure rate D is regarded as the capacity at time interval t of the loop detector departing the segment, which is the flow corresponding to the density threshold μ_{dts} in its FD (Fundamental Diagram), because the speed drops significantly after the density exceeds this value, which means the beginning of the accumulation of queued vehicles. The number of vehicles contained in its adjacent upstream segment is n, i.e. vehicle density, and the number of vehicles contained in it is determined by the state transition probability $\frac{A}{A+D}$ becomes $n+1$, and the state transition probability $\frac{D}{A+D}$ becomes $n-1$. Freeway stretch I-694 contains 8 segments, and except for the first segment, we divide each of the remaining segments into 50 small segments. According to the above Markov process and the density of the previous segment, the vehicle density of the selected segment is obtained, which is equal to the average value of vehicle densities of the 50 small segments it contains, and after simulating 1000 times, the results are averaged, as shown in Fig. I.1.

According to Fig. I.1, we notice that the above Markov process can model the vehicle density spatial distribution well, especially during the busy time of the morning peak period. Therefore, the vehicle distribution on the freeway stretch will satisfy the probability mass described in Eq. (I.1) when the capacity of all segments is assumed to be approximately the same, and the variance of the spatial density distribution also satisfies the Eq. (I.4).

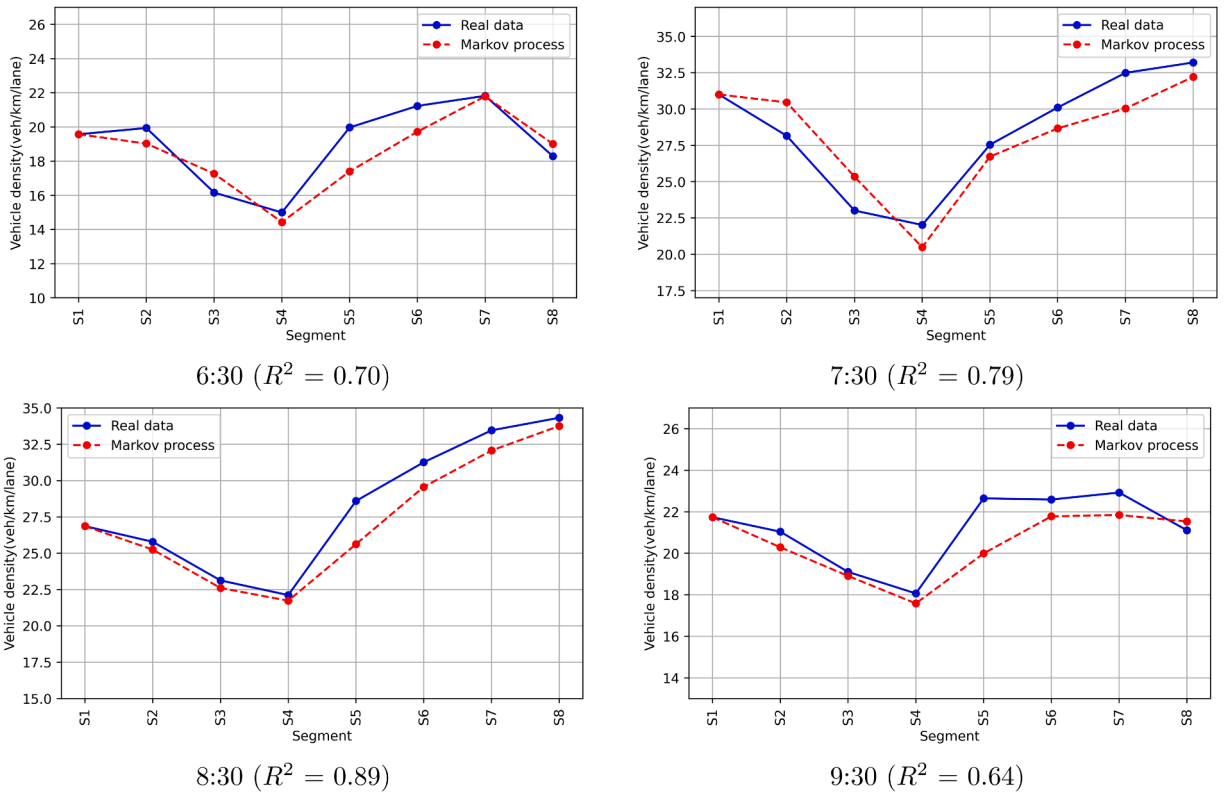


Fig. I.1. The vehicle density distribution results of different time points during the morning peak period obtained by the Markov process simulation.

We know that the variance of the number of vehicles on the freeway stretch σ_n^2 and the variance of the vehicle density of the freeway stretch σ_d^2 satisfy the following relationship.

$$\sigma_n^2 = (L \cdot G)^2 \cdot \sigma_d^2 \tag{I.5}$$

By combining Eqs. (I.4) and (I.5), we obtain the relationship between average vehicle density $\overline{\mu_d}$ and vehicle density heterogeneity σ_d^2 shown in Eq. (I.6).

$$\sigma_d^2 = \frac{D \cdot \overline{\mu_d}^2}{A} \tag{I.6}$$

We use the formula shown in Eq. (I.6) to represent the vehicle density heterogeneity on the freeway stretch. The arrival rate A and the departure rate D are included in Eq. (I.6), where the departure rate D largely depends on the capacity of the freeway stretch itself, and its fluctuation is less than the arrival rate A in time based on the detailed analysis in Section 3.1. Therefore, we treat the departure rate D as the constant and the arrival rate A as the average traffic flow $\overline{A}(t)$ of the freeway stretch.

Based on the model Eq. (I.6) and the real data shown in Fig. 2(b), we estimate the parameter D in Eq. (I.6). The estimated results are 1343.67 veh/hr/lane and 1388.55 veh/hr/lane for freeway stretches I-694 and I-805 respectively. This is consistent with capacity values in the real world. In addition, as indicated by Eq. (I.6), given the same average vehicle density, density heterogeneity decreases with increasing traffic flow, suggesting that congestion is more evenly distributed as vehicle speed increases. This aligns with real-world scenarios, as vehicle speeds near freeflow speed imply an absence of severe bottlenecks on the freeway stretch leading to large differences in vehicle density between upstream and downstream. As a result, vehicles can traverse the freeway stretch freely, causing vehicle density to uniformly distribute across space. This is consistent with the observed reduction in flow throughput due to increased density heterogeneity in the freeway network (Geroliminis and Sun, 2011b). Therefore, we substitute Eq. (I.6) for $\sigma_d^2(t)$ in model Eq. (19), as shown in Eq. (I.7).

$$\sigma_\tau^2(t) = \left(\frac{l \cdot G}{\overline{A}(t)}\right)^2 \cdot \left(\zeta_1 \cdot (J + 2\bar{r} \cdot (J-1)) \cdot (\overline{\mu_d}(t) - \mu_{dts})^2 + \zeta_2 \cdot J \cdot \frac{\overline{\mu_d}(t)^2}{\overline{A}(t)} + \zeta_0\right) \tag{I.7}$$

Where ζ_1, ζ_2, ζ_0, and μ_{dts} are parameters needed to be estimated in the model. The capacity constant D in Eq. (I.6) is embedded in the parameter ζ_2, and therefore does not appear explicitly in the formula.

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